

Deriving the Particle Zoo from Observer Consistency

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Abstract

This paper turns the OPH local pixel fixed point into particle-side structure. A screen cell is read in two ways: as outside area and as an inside electromagnetic observation scale. Matching those readings fixes the pixel used by the particle branch.

With the Standard Model structure supplied by the compact paper, the same setup carries charge quantization, massless photon/gluon/graviton carriers, a fine-structure diagnostic, an electroweak bridge, the hierarchy/naturality result, the Higgs/top split on its stated quantitative branch, selected-class quark/Yukawa support surfaces, and a weighted-cycle neutrino comparison branch. The public numeric particle prediction surface contains the structural carrier zeros and the conditional Higgs candidate on the declared quantitative branch. Target-anchored charged/quark witnesses and compare-only absolute neutrino attachments remain audit or comparison rows. Hadron masses and absolute charged-lepton masses are work in progress.

What This Paper Contributes

The structural particle content is inherited from the compact SM/GR paper: compact gauge reconstruction, the realized Standard Model quotient, exact hypercharge, three colors, three generations, and the massless carrier zeros. This paper asks what that structure does after the local pixel fixed point is turned on.

The OPH addition is the forward particle readout. A single screen-cell coordinate $P_\star$ feeds the electromagnetic branch, the electroweak transport surface, and the mass rows that can be certified on the declared branch. The hierarchy result is the cleanest example: the 12-port screen sieve and the 24-slot oriented repair register give the capacity-electroweak bridge, and the selected exact branch emits the weak scale with Higgs naturality defect $\epsilon_H = 0$. The same paper keeps weaker entries visible without upgrading them. Charged lepton absolute masses and first-principles hadron masses are outside the emitted theorem, benchmark checks are named as checks, and measured endpoint data are isolated when a transport bridge uses them.

1 Introduction

Observer-Patch Holography asks a concrete follow-up question after the gauge structure is fixed. Can one local screen constant organize the particle spectrum, or are the masses and mixings a collection of unrelated inputs?

The structural companion papers start from observer consistency on a finite screen. Physical data are organized in local patch algebras, nearby observers must agree on overlaps, realized states obey a local maximum-entropy and refinement principle, generalized entropy is recoverable, and the realized low-energy branch is fixed by Minimal Admissible Realization (MAR). On the controlled BW/geometric branch, that basis yields the Lorentzian and Einstein branches; on the realized compact-gauge branch it yields the Standard Model gauge structure

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3.$$

The Lorentzian branch, Einstein recovery, and the realized Standard Model gauge structure written above are imported from [5] with their stated branch conditions. That same paper also derives the hypercharge lattice and the realized color triplet $N_c = 3$ and generation count $N_g = 3$. This paper uses those results as input and summarizes only the downstream particle-spectrum continuation. If that derivation is correct, it should also constrain the particle spectrum. The substrate-neutral normal-form framework of Ref. [1] isolates the same-source confluence, cross-source boundary-identification, liveness, and refinement criteria used upstream. It supplies no particle-sector selector or numerical particle output; every such claim below remains conditional on the compact-paper branch and the particle-specific receipts.

The particle branch studied here is a one-fixed-point forward reconstruction problem with an explicit closure matrix. Instead of treating masses, mixings, and Yukawa parameters as independent inputs, it tries to propagate one dimensionless pixel ratio P through the spectrum. The pipeline claims public numeric closure only for the structural massless carriers and the Higgs candidate on its declared quantitative surface. The hierarchy/naturality bridge is closed on its selected branch. The selected-frame quark theorem and weighted-cycle neutrino lane remain visible as support or comparison surfaces, but their target-anchored quark masses and compare-only absolute neutrino attachments are withheld from public prediction tables. The P -closure root, the electroweak W/Z rows, charged leptons, and hadrons have the sector-specific boundary statuses recorded below. This paper develops the derivation chain from the axioms and candidate P trunk to the reported particle outputs, with the closure boundary stated by sector.

1.1 Why this particle content is selected

On the declared branch, particles are not inserted by hand as a list of elementary ingredients. They are the stable observer-visible excitations and carrier modes that are left once three layers of structure have been fixed:

1. overlap consistency on the observer-patch network;
2. compact gauge reconstruction from the theorem-produced transportability criterion, fixed-cutoff bosonic sector category, refinement/fiber ladder, and realized MAR-admissible witness data;
3. minimal admissible realization of the low-energy branch.

The first layer says what kinds of local data can be compared consistently across neighboring patches. The second assembles persistent zero-obstruction sector data into a compact gauge structure through the compact paper's transportability, category, and refinement/fiber theorems. The third selects which of the admissible low-energy branches is physically realized. At that point one is not choosing an arbitrary particle catalog. One is reading off the carrier and matter content of the realized branch.

Recovering Relativity and the Standard Model from Observer Overlap Consistency [5] is therefore central to this draft. That paper does the structural work that fixes the realized low-energy package

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3,$$

together with one admissible Higgs doublet and the realized chiral matter content. The construction does not begin by guessing photons, gluons, quarks, leptons, and weak bosons and then asking how to fit them. It first derives the realized gauge-and-matter branch, and that branch determines which elementary carriers and matter families are available at low energy.

On that selected one-Higgs branch the scalar carrier has a canonical local screen-chart model. Let $C_{\mathrm{EW}} \cong \mathbb{CP}^1$ be the support-visible electroweak chart and fix the positive Hopf line-bundle convention by the neutral component condition $Q(\phi^0) = 0$. Borel–Weil gives

$$H_{\mathrm{OPH}} = H^0(C_{\mathrm{EW}}, \mathcal{O}(1)) \cong \mathbb{C}^2,$$

the first nontrivial holomorphic section space. With OPH’s hypercharge and $\mathbb{Z}_6$ normalization this is exactly the $(1, 2)_{1/2}$ Higgs carrier. A nonzero section direction is a point of $\mathbb{P}(H_{\mathrm{OPH}}) \cong \mathbb{CP}^1$, whose stabilizer is the electromagnetic $\mathrm{U}(1)_Q$. This explains the representation, charge, and symmetry-breaking geometry of the one-Higgs slot. It does not derive m_H , the quartic, v , or Coleman–Weinberg dynamics; the weak scale, Higgs/top quantitative surface, and $\epsilon_H = 0$ hierarchy/naturality closure remain the OPH quantitative branch.

This also explains the phrase “and no others.” The structural OPH claim concerns the realized low-energy branch selected by the stated admissibility conditions. Once one imposes those conditions, the realized low-energy branch is the Standard-Model branch. Additional light gauge bosons, additional light Higgs multiplets, extra light chiral families, or low-energy supersymmetric partners do not appear on that realized branch because they would enlarge the admissible package that MAR selects against. In that sense the observed light particle zoo is not an arbitrary menu; it is the minimal realized one.

The individual families then emerge for different reasons. The photon, gluons, and graviton are structural carriers: once the realized electromagnetic, color, and dynamical-metric branches are present, those massless carriers follow as symmetry-protected zeros. The weak bosons arise when the electroweak gauge sector is propagated through its quantitative closure branch. Quarks and leptons arise from the realized chiral matter package together with the three-generation structure $N_g = 3$. Their family splittings are then read from the deeper overlap-transport and excitation machinery, not from a second arbitrary postulate that says “copy the family three times and assign masses by hand.” Hadrons come one layer later: once color is realized and confined, stable hadrons are composite readout channels of the quark/gluon sector.

The role of the pixel constant P is also important to state clearly. Here, P does not decide whether the photon exists, whether there are three colors, or whether quarks come in three generations. Those structural facts belong to the gauge-and-matter branch fixed before the quantitative closure step. What P does is set the shared quantitative scale on which the realized particle content is read out numerically. So the logic of the paper is:

$$\begin{aligned} \text{observer consistency} &\rightarrow \text{realized gauge/matter branch} \rightarrow \text{particle content} \\ &\rightarrow \text{common scale } P \rightarrow \text{quantitative spectrum.} \end{aligned}$$

That is how the derivation explains why a universe with the realized branch and the shared pixel scale exhibits this particle zoo.

1.2 Logical basis and reading rule

The particle derivation uses the same OPH basis as the SM/GR derivation in Ref. [5]. Its five axioms are the screen net, overlap consistency, local MaxEnt with refinement stability, recoverable generalized entropy, and Minimal Admissible Realization (MAR). On that basis, the SM/GR derivation in Ref. [5] supplies the structural chain used here: compact gauge reconstruction as a classification step, MAR selection of the realized Standard Model quotient, the hypercharge lattice, the realized color triplet $N_c = 3$, and the generation count $N_g = 3$. Refs. [3, 4] supply the patch-net, repair, measurement, and regulated screen language used later when flavor transport and observer-facing readout enter.

The quantitative particle side adds one local closure variable. The common pixel ratio P , fixed on the synthesis paper’s outer/inner closure branch [2], feeds the forward electroweak map and its descendants. The public empirical comparison branch displays

$$\alpha^{-1}(0) = 137.035999177(21), \quad \alpha(0) \simeq 0.00729735256433, \quad P_C \simeq 1.6309682094.$$

The comparison pixel P_C is defined from the measured endpoint; it is not a source-root coordinate. The computation has a fixed source order: golden-ratio entropy balance gives φ , boundary Gaussian normalization supplies the $\sqrt{\pi}$ width, a trial P feeds the source map through unification, running, and electroweak anchoring, and Ward-projected electromagnetic transport gives the Thomson endpoint used by the outer/inner pixel fixed point. The value is forced on that declared branch because each ingredient in the closure equation is fixed by the screen balance, normalization, source map, and Ward-projected endpoint; any different electromagnetic readout gives a different P and fails the outer/inner equality for the same cell. The first-principles diagnostic trunk records

$$\begin{aligned} P &= 1.63097209569432901817967892561191884270169, \\ \alpha_{\text{root}}^{-1} &= 136.994835164621649457949994585787193262029. \end{aligned}$$

The approximately 0.041 inverse-alpha difference to the public Thomson endpoint identifies the declared QCD/hadronic electromagnetic transport boundary; it is not a source-emitted correction. The QCD-free hierarchy witness is the cleaner first-principles stress test. Combining the source/root value with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel, $\alpha_U(P_C)$, gives the mixed-provenance no-hadron diagnostic $A_{\alpha_U}^{\text{fp}} = 137.0359595008\dots$, below the Thomson endpoint. It is comparison bookkeeping rather than a source-only fine-structure prediction. The detailed endpoint table appears in Section 4. The separate cosmic record-closure target $N_{\text{CRC}} = F(N_{\text{CRC}})$, with count representation $N_{\star} = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N]$, belongs to the cosmological-capacity branch and only to legacy capacity-level neutrino side estimates. The Newton coupling uses the separate selected no- G scale certificate $\gamma_{\star} = \ell_{\star} \nu_{\text{Cs}}/c$, equivalently $B_{\star} = 3\pi/\ell_{\star}^2$, so the particle branch does not derive G by back-solving the pixel ratio. The pair $(P_{\star}, N_{\text{CRC}})$ determines dimensionless curvature products such as $\Lambda_{\star} a_{\text{cell}} = 3\pi P_{\star}/N_{\text{CRC}}$, not the SI scale product $\Lambda_{\star} N_{\text{CRC}}$ by itself. The rounded $N_{\text{CRC}} \simeq 3.31 \times 10^{122}$ display is a capacity-scale label, not the high-precision input for G . Structural carriers, quantitative outputs, exact sidecars, and continuation lanes are therefore downstream branches of the declared theorem checklist, with particle-facing quantitative burden carried by the local fixed point $P_{\star}$. The public fine-structure display row is the declared Ward-projected Thomson endpoint of the OPH fixed-point equation. The first-principles diagnostic row and empirical hadron closure rows are separate support records. Section 4 records that boundary in full.

2 P-Closure and the Reverse-Engineering Strategy

Before the claim-tier table, the reverse-engineering claim should be stated plainly. In ordinary phenomenological use, the Standard Model does not derive the observed particle masses and mixings from one common microscopic constant. It is usually presented with 19 free parameters in the minimal massless-neutrino theory, and with at least 26 once neutrino masses and mixing are included, depending on neutrino-sector conventions. The OPH particle derivation compresses that situation into one universal pixel ratio P from the outer/inner closure relation described in the synthesis paper. The same P must drive all downstream bosonic, quark, lepton, neutrino, and hadronic branches.

One clarification matters. The broader zero-input formulation targets the global screen capacity by

$$N_{\text{CRC}} = F(N_{\text{CRC}}),$$

where $F(N)$ is the active cosmic record capacity read back by observers inside the universe supplied with capacity N . The count-density representation is

$$N_{\star} = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N],$$

equivalently, in differentiable form, as the unique stable fixed point of $T_{\eta}(N) = N + \eta d(\log |\Omega_N^{\text{sc}}| - N)/dN$ under the stated strict-concavity certificate. Informally, it is the unique global capacity where the universe reads back its own boundary without deficit or slack, parallel to the local $P_{\star}$ fixed point where outer screen detuning equals the inner Thomson observation scale. The observed branch reads this same capacity as $N_{\text{scr}} \simeq 3.31 \times 10^{122}$. The particle-spectrum derivation studied here uses the local pixel fixed point $P_{\star}$ in place of a long particle-by-particle parameter list.

Observation is allowed to supply a branch hint. That is a consequence of the OPH picture, where the universe is a closed fixed structure and internal observers can read approximate coordinates from experiments. The proof obligation is stricter: the exact $P_{\star}$ used by the particle branch must be computed by the self-referential pixel equation, and the endpoint must be unique on the declared interval. A fitted particle mass can identify a neighborhood. It cannot replace the closure solve.

2.1 Why a single P matters

The structural theorems determine *what kind* of particle world is realized, but not *what absolute scale in GeV* that world occupies. The overlap, modular, gauge, anomaly, and admissibility arguments recover the realized Standard Model branch, the exact hypercharge pattern, three generations, three colors, and the symmetry-protected massless carriers. But those structural results do not by themselves tell us the numerical values of the W boson mass, the Higgs boson mass, or the up-quark mass.

The role of the pixel closure is to supply one common quantitative scale variable for the entire downstream spectrum. In the synthesis paper, the fine-structure lane asks for the nonzero detuning of a holographic screen cell such that the cell's outer geometric displacement from perfect self-similar equilibrium equals the electromagnetic observation scale emitted by the universe living on that same screen. The outer side of the closure is

$$P = \varphi + \alpha_{\text{in}}(P)\sqrt{\pi}.$$

The first-principles computation is a five-step source chain. First, the golden-ratio entropy balance of the local screen cell supplies $\varphi = (1 + \sqrt{5})/2$. Second, boundary maximum-entropy normalization

fixes the width of the leading electromagnetic detuning to $\sqrt{\pi}$. Third, a trial P is sent through the source map

$$M_U(P) = E_P e^{-2\pi P^{1/6}}, \quad E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}},$$

followed by the heat-kernel closure

$$\bar{\ell}_{\text{SU}(2)}(t_2(P)) + \bar{\ell}_{\text{SU}(3)}(t_3(P)) = \frac{P}{4},$$

which selects $\alpha_U(P)$ and the running family $\alpha_i(m_Z; P)$. Fourth, electroweak mixing gives the source anchor

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3}\alpha_1^{-1}(m_Z; P).$$

Fifth, Ward-projected $U(1)_Q$ transport gives the Thomson endpoint

$$A_T(P) = T_Q(a_0(P), F_{\text{src}}(P)) = \alpha_{\text{em}}^{-1}(0; P),$$

and the cell closes when

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

The numerical solve is therefore a root problem for

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}.$$

The branch interval is localized from the observer-facing data, the solver evaluates $A_T(P)$ from the declared source map at each candidate point, and the accepted value is the unique zero of H on that interval. The measured endpoint can locate the interval. It cannot replace the root condition. Thus α is the observer-supporting electromagnetic width of the local pixel, and its value is fixed by the unique solution of this outer/inner consistency equation. The OPH fixed-point readout $\alpha^{-1}(0) = 137.035999177(21)$ gives

$$P \simeq 1.6309682094.$$

The same equation is the local ruler for the downstream particle rows. The five-layer OPH diagnostic trunk emits the source-side diagnostic point

$$\begin{aligned} P &= 1.63097209569432901817967892561191884270169 & , \\ \alpha_{\text{root}}^{-1} &= 136.994835164621649457949994585787193262029 & . \end{aligned}$$

The detailed table records how the diagnostic trunk, endpoint residual, and source-spectral payload fit together. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. Once the fixed point is set, the particle question becomes whether all downstream readouts can be written as

$$X_j = G_j(P)$$

with no new sector-specific constants inserted by hand. This paper studies exactly that downstream map.

2.2 How the electroweak branch works

In the implementation used here, the quantitative electroweak branch is the first major readout from P . Its public target-free output family is

$$\{M_W^{\text{pole}}, M_Z^{\text{pole}}, \alpha_{\text{em}}^{-1}(q^2), \sin^2 \theta_W(q^2), v\},$$

emitted from one declared input P on the printed running/matching/threshold/scheme package. The construction is straightforward in spirit: start from P , build the electroweak running family, select the physical carrier point on that family, and read off the W boson and Z boson pair from that selected point. The Higgs/top critical stage inherits that same electroweak core and does not introduce a new free-input sector. On the declared Ward-projected transport branch, the low-energy electromagnetic row is represented by the Thomson endpoint

$$\alpha_{\text{Th}}^{-1}(P) = \lim_{q^2 \rightarrow 0} \alpha_{\text{em}}^{-1}(q^2; P)$$

of that same electromagnetic transport family. The forward logical order on this branch is:

$$P \mapsto (M_U(P), E_{\text{cell}}(P)) \mapsto \alpha_U(P) \mapsto (t_U(P), t_{\text{tr}}(P)) \mapsto (t_2(P), t_3(P), v(P)) \mapsto \alpha_i(\mu_*; P).$$

The forward transmutation certificate records that the same basis reconstructs the unified diffusion parameter $t_U(P) = 4\pi^2 \alpha_U(P)$ and transmutation exponent $t_{\text{tr}}(P) = 2\pi/((N_c + 1)\alpha_U(P))$ as the pixel-closure solve itself. Measured electroweak data appear only at the validation surface.

The electroweak branch is the place where the common fixed point P_* emits the weak-sector rows. The published W/Z values are

$$M_W = 80.377 \text{ GeV}, \quad M_Z = 91.18797809193725 \text{ GeV}.$$

The selected-carrier chart and the freeze-once coherent repair surface are recorded in the support table below.

2.3 What the electroweak branch fixes

On the declared electroweak running/matching/threshold/scheme surface, the pixel variable P fixes the source basis $(\alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v)$. The target-free first-principles repair theorem fixes the mass-side pair (W, Z) together with the source-locked running-family anchor (a_0, s_0, v) on the declared D10 surface from that source basis fixed by the forward solve. The electromagnetic row is physically read only after Ward projection to the unbroken $U(1)_Q$ channel; its low-energy value is the $q^2 \rightarrow 0$ endpoint of the same transport family. The derivation also includes two explicit benchmark surfaces beneath the public theorem output: the exact selected-carrier chart and the freeze-once coherent repair surface. The same support logic is used later in the paper because flavor continuations have different support levels, whereas the electroweak quantitative-closure lane has an explicit declared-surface contract. Its source payload, same-scheme remainder, and interval-certificate records are public bookkeeping artifacts for the declared bridge from P to the fine-structure endpoint. On the paper surface, the transmutation factor is $\beta_{\text{EW}} = N_c + 1$; overloaded β -ratios appear only on benchmark readouts and are not part of the theorem contract.

3 Results at a Glance

The particle derivation carries the local pixel scale into the directly comparable rows below. Detailed scope statements are collected in the support sections that follow.

Chain	OPH output	Role
P -closure and fine structure	$\alpha_{\text{root}}^{-1} = 136.9948351646 \dots$; $A_{\alpha_U}^{\text{fp}} = 137.0359595008 \dots$; endpoint $\alpha^{-1}(0) = 137.035999177(21)$	source/root audit, mixed-provenance comparison diagnostic, and measured endpoint
Structural massless carriers	$m_\gamma = m_g = m_{\text{grav}} = 0$	symmetry-protected zeros
Electroweak W/Z	80.377 GeV 91.18797809193725	weak-sector benchmark pair
Electroweak hierarchy/naturality	$N_{\text{CRC}}^{\text{EW}} \simeq 3.53235 \times 10^{122}$ $\mathcal{B}_{\text{EW}} = 0, \epsilon_H = 0$	selected local/global bridge
Higgs/top relation	$m_H = 125.1995304097179$ GeV, with a companion top coordinate on the same split surface	conditional declared split surface; top coordinate is not a separate public top-mass prediction row
Charged leptons	target-anchored witness values withheld	empirically anchored same-family audit witness
Selected-frame quarks	target-anchored running sextet withheld; exact forward Yukawa support surface retained	selected public quark-frame audit/support surface
Neutrino absolute attachment	compare-only absolute masses withheld; scale-free and mixing comparison rows retained	weighted-cycle comparison branch
Hadrons	first-principles strong-binding descent plus empirical $e^+e^- \rightarrow$ hadrons closure surface	OPH hadron construction for first-principles masses; measured hadron data for empirical closure rows
H3 record localization and worldlines	certified cap-response localization balls followed by cross-boundary record-token continuation, or AMBIGUOUS	conditional observer-record theorem; not a species, mass, or coupling input

Two points are worth stating explicitly. First, the theorem rows and the exact-hit rows are not the same object. The theorem table carries selected-frame quark closure and a corpus-limited charged no-go boundary, while the exact-hit surface above it also contains exact same-family charged and quark witnesses, the restricted transport-frame quark exact chain, and exact diagnostic sidecars. Second, “the bosons are finished” is too blunt. The photon, gluons, and graviton are structural zeros; the W boson and Z boson are benchmark frozen-adaptor values; and the Higgs boson sits on the declared Higgs/top critical surface.

This support boundary governs the rest of the paper. Whenever a later chapter gives more detail about a continuation family, it should be read through this table. The table records the constructive chain and keeps its support status explicit.

Exact non-hadron hit surface. If one asks only for exact hits, the paper-facing non-hadron bundle splits by lane as follows:

Concretely, the public values and benchmark checks shown on this surface are

$$\alpha^{-1}(0) = 137.035999177(21), \quad P \simeq 1.6309682094,$$

$$N_{\text{CRC}}^{\text{EW}} = 3.5323546226929906511 \dots \times 10^{122}, \quad \mathcal{B}_{\text{EW}} = 0, \quad \epsilon_H = 0,$$

$$(M_W, M_Z) = (80.377, 91.18797809193725) \text{ GeV}$$

as a frozen benchmark adaptor, and

$$m_H = 125.1995304097179 \text{ GeV}$$

as the conditional Higgs candidate on the declared D10/D11 surface. Target-anchored charged-lepton and quark witness values, including the selected-class running-quark sextet, are retained

only in the exact-fit audit artifacts. Compare-only absolute neutrino attachments are likewise withheld from public prediction tables. The neutrino comparison surface keeps representative splitting checks such as

$$\begin{aligned}\Delta m_{21}^2 &= 7.488059465106851 \times 10^{-5} \text{ eV}^2, \\ \Delta m_{31}^2 &= 2.5123118727618473 \times 10^{-3} \text{ eV}^2, \\ \Delta m_{32}^2 &= 2.4374312781107786 \times 10^{-3} \text{ eV}^2.\end{aligned}$$

The table is lane-based. It states the derivation chain that emits each exact-hit surface and the caveat that prevents a stronger theorem claim.

How to read mismatches and exact hits. Exact numerical agreement is not by itself a proof of a blind prediction. The provenance record classifies the W/Z row as target-used frozen-reference reproduction, the charged-lepton triple as an empirically anchored current-family witness, and the quark sextet as a target-anchored selected-frame witness rather than a public source-only prediction. The same record classifies the hierarchy/naturality bridge as a selected-branch source-side theorem with zero bridge residual and $\epsilon_H = 0$. Where a row does not match, or where it matches only on a constrained surface, the explanations are as follows. The P -closure and $\alpha(0)$ displacement is localized to the Ward-projected Thomson residual. At the implemented fixed point the residual is

$$0.041164012378350542050005414212806738$$

inverse-alpha units. Combining the undressed source/root inverse coupling with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\text{fp}} = 136.99483516462165 \dots + 0.041124336195630495 = 137.03595950081728 \dots,$$

The sum uses $\alpha_U(P_C)$, while the root is emitted at P_{cand} . It lands close to the Thomson endpoint and does not reach it, but its mixed provenance prevents source-only promotion. OPH source-only closure requires the Ward-projected hadronic spectral transport. The remaining inverse-alpha shortfall is

$$137.035999177 - A_{\alpha_U}^{\text{fp}} = 0.000039676182720047050005414212807 \dots$$

Equivalently, the calibrated fixed-point residual can be written

$$\Delta_{\text{H,cal}}^{\text{fp}} = \alpha_U(P_C) C_{24,Q}, \quad C_{24,Q} = 1.0009647859732326253849511140702475 \dots$$

The $C_{24,Q}$ factor is endpoint accounting rather than a source-emitted theorem. At the public endpoint pixel value

$$P \simeq 1.6309682094$$

the endpoint residual package requires

$$\Delta_{\text{source}}(P) = 0.041465861005223389053448715357314044 \dots$$

That scalar belongs to the source-side hadronic spectral transport and scheme-remainder map. The declared empirical route would use a separately labeled $e^+e^- \rightarrow \text{hadrons}$ spectral input. No same-scheme integrated endpoint payload is emitted here. The first-principles transport row excludes an inserted comparison endpoint. The electroweak support record names the residual map and the RG/matching/threshold/scheme packet. An observer inside the branch measures the dressed

Thomson coupling, not the undressed source diagnostic $1/136.994835\dots$, because a zero-momentum electromagnetic measurement sees the Ward-projected $U(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme finite endpoint matching have been included. The electroweak W/Z row is therefore a benchmark check. Charged leptons carry a corpus-limited no-go because the determinant trace-lift attachment from the D10 descendants of P to physical charged data is absent. The auxiliary direct-top PDG row differs from the theorem coordinate by 0.20673301656674425 GeV, or 0.28458848947515303 combined standard deviations, because Q007TP and Q007TP4 are distinct extraction codomains; the direct-top response kernel has a corpus-limited no-go boundary. Neutrino absolute masses are not directly measured, and PMNS-angle residuals remain visible comparison tension outside the theorem branch. First-principles hadron masses have no emitted prediction because the required Ward-projected hadronic spectral measure must come from a working OPH hadron construction. Empirical hadron closure rows carry a separate support class.

Local unification surface. The bosonic rows carry one cross-lane statement. The same pixel input P , fixed on the synthesis-paper outer/inner closure relation, fixes the D10/D11 bosonic trunk

$$P \mapsto \alpha_U(P) \mapsto (t_U(P), t_{\text{tr}}(P)) \mapsto v(P) \mapsto (M_W, M_Z).$$

$$\sigma_{D11, \text{HT}} = \alpha_U(P) \cos(2\theta_{W0})/\sqrt{\pi} \mapsto (m_H, m_t).$$

Here $\alpha_U(P)$ is the branch value selected by the same forward D10 pixel-closure solve, so the bosonic trunk contains no inverse electroweak readback of the internal transmutation data. On the companion gravity side, the same pixel law packages

$$\bar{\ell}_{\text{SU}(2)}(t_{2,\text{run}}) + \bar{\ell}_{\text{SU}(3)}(t_{3,\text{run}}) = P/4.$$

The gravity normalization itself is emitted by the selected scale certificate $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. With $a_{\text{cell}} = P\ell_\star^2$, the Newton area law gives $G_{\text{geom}} = \ell_\star^2$, so the pixel cancels. The local unification surface separates one explicit familiar-unit readout package from three support surfaces: the W/Z benchmark sidecar, the exact Higgs/top theorem surface, and the gravity-side release with its strict classical-regime clause. On the gravity side, the stated local extension surface uses the lifted product presentation of the realized quotient branch and identifies

$$\bar{\ell}_{\text{shared}} = \bar{\ell}_{\text{SU}(2)}(t_{2,\text{run}}) + \bar{\ell}_{\text{SU}(3)}(t_{3,\text{run}}).$$

On that same surface the D10 pixel law fixes $\bar{\ell}_{\text{shared}} = P/4$, and the local SI readout is

$$G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}$$

from the selected scale certificate. The χ_ν protected-reserve branch uses the same public endpoint convention as

$$P_\chi = P_C = 1.630968209403959\dots, \quad \epsilon_{\text{res}} = \frac{P_\chi}{24},$$

after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass. This does not make $P/4$ a primitive Hilbert-space dimension: $P/4$ is the local screen-cell entropy budget in the particle and hierarchy lanes, while $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch. On that declared extension surface the same familiar-unit package reads

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}} \hat{L}(P), \quad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c} \hat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c}{\sqrt{a_{\text{cell}}}} \hat{E}(P), \quad \Theta_{\text{loc}} = \frac{\hbar c}{k_B \sqrt{a_{\text{cell}}}} \hat{\Theta}(P),$$

with dimensionless $\hat{L}, \hat{T}, \hat{E}, \hat{\Theta}$. Thus, at fixed P , the local ruler is $\sqrt{a_{\text{cell}}}$; seconds are that ruler divided by the structural Lorentz output c ; and GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B . On that declared extension surface the local release bundle is

$$c = 299792458 \text{ m/s},$$

$$G = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2},$$

$$(M_W, M_Z, M_H) = (80.377, 91.18797809193725, 125.1995304097179) \text{ GeV},$$

where M_W and M_Z are benchmark frozen-adaptor values and M_H is the declared Higgs/top-surface value. The paper keeps c as the structural Lorentz output and keeps the G row as an exact scale-readback value, with no literal zero-difference identity against the rounded benchmark 6.6743×10^{-11} .

The detailed inventory below records the public theorem/continuation rows sector by sector. Whenever an exact sidecar or same-family witness is stronger than the public theorem row, the caveat is the one stated in the exact-hit table above.

3.1 Detailed particle inventory

Family	Particle	Support status	OPH value or strongest derived benchmark	Remaining step
Structural carriers	photon	structural	0	none
Structural carriers	gluons	structural	0 for the color gauge sector	none
Structural carriers	graviton	structural	0 for the dynamical-metric spin-2 carrier	none
Electroweak branch	W boson	benchmark adaptor	80.377 GeV on the frozen W/Z validation surface	benchmark check beneath the declared P -to-fine-structure arithmetic bridge; public validation records are source spectral measure payload, same-scheme remainder, and interval certificate
Electroweak branch	Z boson	benchmark adaptor	91.18797809193725 GeV on the frozen W/Z validation surface	same benchmark caveat as the W row
Higgs/top critical stage	Higgs boson	quantitative theorem	125.1995304097179 GeV on the declared running, matching, and threshold surface	exact first-principles Higgs/top split theorem; benchmark inverse Higgs/top slice is on the same Jacobian surface
Quark family	top quark	selected-frame audit witness	withheld from public prediction table	target-anchored selected-class running-mass witness on f_P ; the split lane carries a companion top coordinate on its declared quantitative surface, but no public top-mass prediction row is promoted
Quark family	up quark	selected-frame audit witness	withheld from public prediction table	target-anchored selected-class running-mass witness on f_P ; global classification of all public quark frame classes is closed as a corpus-limited no-go
Quark family	down quark	selected-frame audit witness	withheld from public prediction table	same selected-frame audit surface; target-free mass bridge closes only on the emitted mass ray
Quark family	strange quark	selected-frame audit witness	withheld from public prediction table	same selected-frame audit surface; the lower selector is a negative sheet statement beneath that theorem
Quark family	charm quark	selected-frame audit witness	withheld from public prediction table	same selected-frame audit surface; exact forward Yukawa support data are retained on f_P
Quark family	bottom quark	selected-frame audit witness	withheld from public prediction table	same selected-frame audit surface; the exact sextet matches the official PDG 2025 API running-quark target surface because the physical sigma datum is target-derived

Family	Particle	Support status	OPH value or strongest derived benchmark	Remaining step
Charged leptons	electron	continuation gap	n/a ; exact same-carrier centered readback exists once a charged source pair is emitted, but the absolute scale is blocked by the closed common-shift no-go; benchmark target $g_e^* = 0.0457789$, equivalently $\Delta_e^{\text{abs},*} = 3.00398633$	close branch-generator splitting, then emit the lift whose descended scalar is $\mu_{\text{phys}}(Y_e)$; within that lift the exact smaller forcing object is the physical identity-mode equalizer, after which $\tilde{C}_e(Y_e) = \hat{C}_e(Y_e) + \mu_{\text{phys}}(Y_e) \mathbf{1}$, $A_{\text{ch}}(Y_e) = \mu_{\text{phys}}(Y_e)$, and the readouts g_e , Δ_e^{abs} follow
Charged leptons	muon	continuation gap	n/a ; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\hat{C}_e^{\text{cand}} \rightarrow$ branch-generator splitting $\rightarrow$ lift $\rightarrow \mu_{\text{phys}}(Y_e) \rightarrow A_{\text{ch}} \rightarrow g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Charged leptons	tau lepton	continuation gap	n/a ; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\hat{C}_e^{\text{cand}} \rightarrow$ branch-generator splitting $\rightarrow \mu_{\text{phys}}(Y_e) \rightarrow A_{\text{ch}} \rightarrow g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Neutrino comparison	declared f -basis weighted-cycle matrix	rejected target-informed candidate	no flavor-particle mass row; the frozen declared-basis coordinate gives $\theta_{12} = 34.2259^\circ$, $\theta_{23} = 49.7228^\circ$, $\theta_{13} = 8.68636^\circ$, $\delta = 305.581^\circ$, $J = -0.02753$, and, under its declared normal-ordering labels, $\Delta m_{21}^2 / \Delta m_{32}^2 = 0.03072111$	the upstream flavor kernel is a hand-written template; the physical charged basis and mass-label rule are open; target-ranked selector development prevents historical-blindness status; the NuFIT 6.1 correlated profile rejects the stored candidate; Majorana and absolute-attachment values remain compare-only
Hadrons	proton	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	first-principles prediction requires a working OPH-QCD hadron backend, such as GLORB/Echosahedron, plus Ward-projected two-current, higher-point, and transition spectral exports, same-scheme remainder, and production systematics; empirical closure rows use separate $e^+e^- \rightarrow$ hadrons input
Hadrons	neutron	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate, with isospin-resolved hadron systematics in the construction support class
Hadrons	neutral pion proxy	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate; local stable-channel surrogate output is not a paper prediction
Hadrons	$\rho(770)^0$ proxy	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate, plus finite-volume resonance extraction in the construction support class

4 Detailed Closure Values, Screen Architecture, and Theorem Packages

This section records the detailed closure-value and theorem checklist used by the particle derivation: the five axioms, the screen-capacity closure, the local pixel closure, the regulated screen realization, and the imported OPH theorem packages. The order is the logical order used later in the paper: axioms and screen language first, closure values next, and theorem packages after that.

4.1 Canonical OPH basis

The particle-spectrum derivation uses the same five axioms as the compact reconstruction paper [5]. They are restated here for local readability because every later particle-family chapter depends on them either directly or through the structural OPH chain imported from that paper.

Axiom 4.1 (Screen Net). *Physical data is organized on a horizon screen S^2 carrying a net of local*

algebras

$$P \mapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 4.2 (Overlap Consistency). *For overlapping patches $P_1 \cap P_2 \neq \emptyset$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 4.3 (Local MaxEnt and Refinement Stability). *At the regulator scale ℓ_{UV} , the realized branch is selected by maximizing entropy subject to a finite family of gauge-invariant local constraints*

$$\mathcal{C}_{\ell_{UV}} = \{O_a(x)\}_{a=1}^{N_{\text{con}}},$$

with support radius $O(\ell_{UV})$ and with label set independent of the number of regulator cells. Under refinement, the same finite constraint family persists, so the realized states belong to one common finite-dimensional MaxEnt family instead of unrelated maximizers.

Axiom 4.4 (Recoverable Generalized Entropy). *A generalized entropy functional exists on caps,*

$$S_{\text{gen}}(C) = S_{\text{bulk}}(C) + \langle L_C \rangle,$$

where L_C is a positive edge-center entropy functional and the semiclassical branch identifies its leading coarse-grained contribution with $A(\partial C)/(4G)$. The functional obeys the recoverability and focusing structure required by the collar and null-modular arguments.

Axiom 4.5 (Minimal Admissible Realization). *Among admissible realized low-energy sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the EFT branch, its admissible light chiral matter content, and one Higgs doublet, the realized package is lexicographically minimal under*

$$C(\mathfrak{S}) = (\chi_{\text{cpl}}, N_{\text{nonab}}, N_c, N_g),$$

subject to loop coherence, anomaly freedom, refinement-stable light chiral matter, single-Higgs Yukawa completable structure with one connected abelian charge factor, intrinsic CP capability, and weak-sector UV completeness.

For particle physics, the first four axioms provide the observer-centric kinematic and entropic/modular background. The fifth axiom, MAR, is the selector that turns “some compact gauge group reconstructed from the transportable edge-sector category under the compact paper’s explicit coherent refinement ladder, symmetry, and finite-fiber conditions” into the realized Standard Model branch. This paper therefore uses MAR constantly, but only after the compact-gauge reconstruction step has been separated from the later realized-branch selection.

4.2 Regulated screen architecture and patch language

The SM/GR derivation in Ref. [5] states the axioms in algebraic language. Ref. [4] supplies one concrete regulated realization of that language: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap packets, record algebras, and local repair instruments. A spherical screen is an observer-visible regulator chart, not a literal microscopic computer. This regulated architecture is important for the particle derivation for three reasons.

First, it makes the screen picture operational. A patch is a finite local algebra in a bounded carrier, and an overlap is a boundary-visible algebra with declared shared observables. Second, it makes the edge-sector language concrete. The particle derivation uses edge sectors, transport, refinement, fusion, and overlap data to build the gauge and flavor branches; Ref. [4] shows how those objects can be realized at fixed cutoff by finite gauge-aware patch carriers. Third, it clarifies the measurement interface. The observer-facing measurement package is a theorem-bearing fixed-cutoff statement about a central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting algebra.

This regulated architecture should be read as a reference implementation, not as the unique OPH UV completion. Ref. [4] is explicit on that boundary: it selects a federated patch-carrier architecture because it is local, finite-dimensional, gauge-aware, naturally compatible with collars, overlap observables, and repair maps, and does not ask the reader to identify the spherical regulator chart with the literal substrate. Hardware evidence is imported only when it appears in a public OPH evidence bundle with stable hashes and verifier receipts. This paper inherits that architecture as a concrete screen-side realization of the OPH language, while keeping microscopic uniqueness separate from the support-visible continuum theorem surfaces supplied by the compact paper.

4.3 Quantitative inputs and where they enter the particle derivation

The quantitative derivation developed here uses two closure coordinates together with the selected no- G scale certificate:

$$P \equiv a_{\text{cell}}/\ell_{\star}^2, \quad (1)$$

$$N_{\text{CRC}} \equiv \log \dim \mathcal{H}_{\text{tot}}, \quad (2)$$

$$\gamma_{\star} \equiv \frac{\ell_{\star} \nu_{\text{Cs}}}{c}, \quad B_{\star} \equiv \frac{3\pi}{\ell_{\star}^2}. \quad (3)$$

The particle-spectrum draft needs to keep these closure values visible because several later sections depend on them in different ways.

Pixel area P . The pixel area is the declared local UV-area ratio. In this derivation it feeds the heat-kernel / edge-law input stage and, through that route, the forward D10 electroweak surface. It is the common upstream numerical variable for the electroweak, Higgs/top, flavor, quark, and hadron-facing branches. The particle paper therefore treats $P = P_{\star}$ as the inherited outer/inner fixed point

$$P_{\star} = \varphi + \frac{\sqrt{\pi}}{A_T(P_{\star})}$$

from the synthesis paper [2], not as a quantity derived again inside the particle-spectrum argument itself.

Screen capacity N_{CRC} . The screen capacity is not part of the local recovered-core gravity/gauge derivation. It enters the separate screen-capacity branch through the cosmic record-closure target

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \quad \Lambda_{\text{CRC}} \ell_{\star}^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last equality is the scale-certified display. It does not determine an SI curvature scale from $P_{\star}$ and N_{CRC} alone. The count-density representation is

$$N_{\star} = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N].$$

In differentiable form, with $\ell(N) = \log |\Omega_N^{\text{sc}}| - N$, this same branch is the unique stable fixed point of $T_\eta(N) = N + \eta \ell'(N)$ when the OPH normal-form count gives a strictly concave normalized density. Thus the capacity is the unique OPH-derived global readback balance point, not a separately fitted cosmological number. In the particle context, this matters mainly for a legacy capacity-level neutrino side estimate and for the cosmological-capacity discussion that frames why local null data do not determine the cosmological constant. The implemented branch uses the static-patch normalization

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P} \right)^2 \simeq 1.05 \times 10^{122}, \quad N_{\text{scr}} = \pi N_{\text{patch}} \simeq 3.31 \times 10^{122}.$$

This paper should therefore keep the neutrino chapter direct: legacy capacity-level order-of-magnitude bookkeeping and the weighted-cycle neutrino derivation are not the same thing.

Selected scale certificate. The gravity scale is a separate observation-located certificate. On the declared branch,

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \frac{3\pi}{\ell_\star^2}, \quad G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

The local pixel $P_\star$ then identifies $a_{\text{cell}} = P_\star \ell_\star^2$ and $\bar{\ell}_{\text{shared}} = P_\star/4$. In the Newton area law $P_\star$ cancels, so this paper does not treat G as a particle-sector consequence of $P_\star$. $\bar{\ell}_{\text{shared}}$ is a shared-cut density, not the logarithm of an autonomous cell Hilbert factor. The particle branch uses it for cell/edge consistency; it does not select a fixed primitive observer capacity.

Why these are the declared continuous closure/readback values. One of the discipline constraints of the OPH derivation is that particle-sector freedom should not be hidden in a large uncontrolled parameter set. This paper exposes $P_\star$, the N_{CRC} readback fixed point with $N_\star$ as its count representation, and the selected scale certificate, then asks the overlap, modular, gauge, and admissibility machinery to do the rest. Whether later continuation branches stay faithful to that discipline is one of the central questions this paper is meant to make explicit. For the particle-spectrum branch specifically, the nontrivial common quantitative burden sits on $P_\star$: N_{scr} enters the separate capacity/cosmology side and $\gamma_\star$, equivalently $B_\star$, enters the separate gravity-scale side, whereas the reverse-engineering claim for masses and couplings is that one shared pixel fixed point should drive the spectrum instead of a long sector-by-sector input list.

4.4 Technical theorem data carried into the particle-spectrum derivation

This paper uses the same theorem checklist as *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5], together with the regulator-language clarification integrated into the consensus and microphysics appendices. The ledger matters only insofar as it separates branch-internal statements from declared inputs.

The third axiom internalizes several pieces of infrastructure that often appear as separate assumptions: the local Gibbs form of the regulator-scale state, the quasi-local propagation bound, the endpoint-control estimate for bounded intervals, and the meaning of refinement stability itself. In regulator language, one works with finite local Hilbert spaces and a boundary-fixed overlap action on cut data, often summarized as the R0/R1 presentation. Beyond that regulator package, the compact paper supplies the support-visible BW scaling theorem for the Lorentz/null-modular/Einstein branch. Transportability, the fixed-cutoff bosonic sector category, the refinement/fiber ladder, and the realized MAR-admissible compact-gauge witness data are theorem-produced in the compact

paper. On the central branch the matching gluing obstruction is the zero-obstruction transport criterion, not a second theorem-side input. Any local-Gibbs or collar-mixing language on that branch is fixed-cutoff recoverability/support control; it is not the source of the compact-gauge witness. For the BW/geometric side, the target is the support-visible extracted prime geometric subnet. The compact paper proves the needed scaling theorem by combining regularized support-visible modular transport, weak-*/GNS extraction of the cap pair, support-readable modular covariance, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π -KMS convergence with wrong-scale controls. The unregularized full-algebra common-floor route is deliberately not claimed, because off-support directions may collapse without affecting observer-facing matrix elements. Bare finite consensus does not by itself produce the finite cap-normal BW certificate used by that theorem.

For the particle-spectrum derivation, this has a practical consequence. Structural statements such as compact-gauge reconstruction, the realized Standard Model quotient, the exact hypercharge lattice, the color count $N_c = 3$, the generation count $N_g = 3$, and the symmetry-protected massless carrier zeros are not floating independently of the declared input and theorem checklist. They live on a controlled chain whose scope conditions need to be visible in this paper, especially once later chapters turn from structural branches to quantitative closure, continuation, and nonperturbative sectors.

4.5 Core theorem packages used later

This paper will repeatedly use earlier OPH results without re-proving them in full. The theorem packages imported into the present draft are the following.

1. **Patch-net and overlap language.** Ref. [3] organizes overlap repair, fixed-point language, cycle holonomy, and gauge-as-gluing in a form that is especially useful for flavor transport and observer-centric interpretation.
2. **Screen-side regulated architecture.** Ref. [4] gives the federated echosahedral patch-carrier model, the regulated patch-net embedding theorem, and the fixed-cutoff edge heat-kernel / Casimir theorem.
3. **Measurement interface.** Ref. [4], together with the integrated measurement appendices and the synthesis paper *Observers Are All You Need* [2], supplies the fixed-cutoff central-record, Born-rule, and Lüders-conditioning package together with the fixed-cutoff Bell/CHSH theorem stack used when later chapters discuss how measurements pick out definite observer-accessible outcomes and how two-wing comparison laws stay on the quantum side of the classical Bell bound.
4. **Compact gauge reconstruction and Standard Model structure.** *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5] and the dedicated gauge-group fragment together provide the path from refinement-stable edge sectors to a compact group, then from MAR to

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_g = 3, \quad N_c = 3.$$

5. **Continuation-level topic notes.** The integrated lane appendices carry useful material on supersymmetry, heuristic baryogenesis continuations, proton stability, generation structure, and Yukawa hierarchy, but those sections have to be read with their stated claim boundaries

intact. They are sources for later discussion and continuation chapters, not automatic upgrades to recovered-core theorems.

This reading rule governs the whole paper. Structural gauge and carrier results can be used as structural results. The electroweak branch is a forward-emitted Phase-II quantitative branch. The Higgs/top critical stage is a source-only split quantitative theorem on its declared running, matching, and threshold surface. The quark, charged-lepton, neutrino, and hadron chapters retain their stated continuation or comparison labels unless the underlying derivation chains close more strongly.

5 Gauge Architecture, Standard Model Structure, and Structural Carriers

The particle-spectrum derivation does not start from a list of particles. It starts from the compact-gauge branch. OPH reconstructs a compact gauge group from the zero-obstruction edge-sector category produced by the compact paper’s transportability, fixed-cutoff category, and refinement/fiber theorems. The overlap/transport obstruction calculus is the classification stage, not the selection stage. The compact-gauge witness theorem supplies nonempty realized MAR-admissible branch data, and MAR then selects the realized low-energy package. That logical order matters for this paper. The gauge branch fixes the structural carrier content before any mass readout, and it also explains why some rows are exact structural zeros instead of fitted or accidentally small numbers.

On the realized branch, the gauge/content result used throughout this paper is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_g = 3, \quad N_c = 3.$$

Recovering Relativity and the Standard Model from Observer Overlap Consistency [5], the longer main derivation surface, and the dedicated gauge-group proof fragment all agree on the logical split: compact-gauge reconstruction first, realized Standard Model selection second. The first stage gives a compact internal symmetry group under the stated zero-obstruction bosonic sector conditions. The second stage uses MAR and anomaly cancellation to fix the realized quotient and hypercharge lattice; the minimal coupled carrier fixes $N_c = 3$; intrinsic CP capability together with weak-sector UV completeness bounds N_g ; MAR then fixes $N_g = 3$; and Witten parity remains a consistency check on the realized triplet-doublet package.

For the particle zoo this structural branch fixes three exact carrier statements. First, the photon is massless on the realized electromagnetic branch. On the same branch, the abelian curvature and current obey the Maxwell equations after canonical electromagnetic normalization. Second, the gluons are massless on the color branch, even though free gluons are never seen as asymptotic particles because color is confined. Third, the graviton is massless on the dynamical-metric branch of the gravity chain. These are symmetry-protected zeros of the realized branch, not continuation-level masses.

These structural zeros should not be confused with the later quantitative particle rows. The paper should therefore treat the photon, gluons, and graviton as consequences of the gauge and gravity branches themselves, not as outputs of the later quantitative or flavor machinery. The later boson sections will add the W boson, the Z boson, and the Higgs boson, but those are not structural zeros; they sit on electroweak and Higgs/top quantitative stages.

6 Electroweak D10 Branch and the Higgs/Top Critical Stage

The reported bosonic sector distinguishes sharply between emitted numerical rows and scoped non-emitted rows. The active bosonic stages are the electroweak D10 prediction stage and the Higgs/top critical stage, while charged-lepton rows carry their no-go boundary, the neutrino lane contains a rejected weighted-cycle comparison candidate, and hadron rows are backend-gated. The bosonic numerical sector is therefore concentrated in two places: the D10 electroweak branch for the W boson and Z boson, and the Higgs/top critical stage for the Higgs boson together with the exact D11 split-theorem top value.

6.1 Electroweak prediction stage on the D10 quantitative-closure branch

The electroweak derivation consists of a single- P running family followed by a reduced two-scalar carrier, a selector on that carrier, an exact carrier mass chart, and then a target-free source-only transport theorem beyond the selected carrier. In practical terms, the construction starts from the declared pixel input P , builds the running electroweak family, reduces it to the selected carrier, reads the W boson/ Z boson pair from the selected point on that carrier, and then emits the mass pair together with the Ward-projected electromagnetic transport family from the source basis alone. On the D10 branch, a live forward prediction must respect exactly that ordering: first certify the shared pixel input P on the declared D10 running/matching/threshold/scheme surface, thereby fixing the source basis

$$(\alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v),$$

with $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, fixed by the same forward pixel-closure solve, and only then emit the electroweak transport family forward from that source basis. The completion pipeline classifies that full chain outside the promoted theorem tier. Its W/Z rows are the frozen validation adapter values, and their exact numerical agreement is recorded as a comparison label instead of as a closed D10 Phase-II prediction theorem.

The same quantitative lane has one exact golden-ratio benchmark. If one writes

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)}$$

for the total/bulk/edge hierarchy and imposes exact self-similar balance

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle},$$

then x obeys $x^2 - x - 1 = 0$, so the unique positive equilibrium point is

$$x = \varphi := \frac{1 + \sqrt{5}}{2}.$$

Equivalently the order parameter $A_\varphi(x) = x - 1 - \frac{1}{x}$ vanishes there. The synthesis paper fixes the exact numerical value of P through the outer/inner closure relation. The point of the equilibrium theorem here is that the proximity to φ has a structural reason. Exact balance is too symmetric to support durable records, structure, and dynamics, so the closure solve fixes a small equilibrium-breaking detuning away from that balance point.

The compare-only W/Z adapter rows in the final bundle are

$$m_W = 80.377 \text{ GeV}, \quad m_Z = 91.18797809193725 \text{ GeV}.$$

These numbers sit on the frozen validation surface, not on a target-free prediction theorem. The exact selected-carrier chart is explicit on disk and emits the local pair

$$m_W^{\text{carrier}} = 80.38629169244275 \text{ GeV}, \quad m_Z^{\text{carrier}} = 91.18290444674243 \text{ GeV},$$

so the mathematics distinguishes three electroweak objects: the exact selected-carrier chart, the freeze-once coherent repair surface, and the target-free source-only theorem surface that would supply public prediction rows once the P , Thomson, and RG/matching/threshold/scheme gates are closed.

Theorem 6.1 (Freeze-once coherent electroweak repair law). *Let the emitted running/core electroweak basis be*

$$Q_{\text{run}}(P) = (\alpha_{Y,m_Z}(P), \alpha_{2,m_Z}(P), v(P), \eta_{\text{source}}(P)),$$

and write

$$\tau_Y(\tau_2) = -\frac{\tau_2 + 2\eta_{\text{source}}}{1 + 4\tau_2^2}, \quad n_{\text{fiber}}(\tau_2) = 1 + \frac{\alpha_{Y,m_Z}\tau_Y(\tau_2) + \alpha_{2,m_Z}\tau_2}{\alpha_{Y,m_Z} + \alpha_{2,m_Z}}.$$

Freeze one authoritative target pair

$$\mathcal{T}^\dagger = (M_W^\dagger, M_Z^\dagger), \quad 0 < M_W^\dagger < M_Z^\dagger,$$

and define the charged anchor

$$\tau_{2,W}^\dagger = \frac{M_W^{\dagger 2}}{\pi v^2 \alpha_{2,m_Z}} - 1, \quad \delta\alpha_2^\dagger = \frac{M_W^{\dagger 2}}{\pi v^2} - \alpha_{2,m_Z}.$$

Then the fiber-parallel hypercharge leg is

$$\tau_Y^\dagger = \tau_Y(\tau_{2,W}^\dagger), \quad n_{\text{fiber}}^\dagger = n_{\text{fiber}}(\tau_{2,W}^\dagger),$$

$$M_{Z,\text{fiber}}^\dagger = v\sqrt{\pi(\alpha_{Y,m_Z} + \alpha_{2,m_Z})n_{\text{fiber}}^\dagger},$$

and the orthogonal neutral-shear scalar is

$$\delta M_Z^\perp = M_Z^\dagger - M_{Z,\text{fiber}}^\dagger,$$

equivalently

$$\delta n^\dagger = \frac{(M_Z^\dagger + M_{Z,\text{fiber}}^\dagger)\delta M_Z^\perp}{\pi v^2(\alpha_{Y,m_Z} + \alpha_{2,m_Z})}, \quad \delta\alpha_Y^\perp = \frac{(M_Z^\dagger + M_{Z,\text{fiber}}^\dagger)\delta M_Z^\perp}{\pi v^2}.$$

The fiber-parallel hypercharge motion is

$$\delta\alpha_Y^\parallel = \alpha_{Y,m_Z} \frac{8\eta_{\text{source}}(\tau_{2,W}^\dagger)^2 - \tau_{2,W}^\dagger}{1 + 4(\tau_{2,W}^\dagger)^2}.$$

Therefore the unique coherent repair package is

$$\Sigma_{EW}^\dagger = (\delta\alpha_2^\dagger, \delta\alpha_Y^\parallel, \delta\alpha_Y^\perp),$$

equivalently $(\tau_{2,W}^\dagger, \delta n^\dagger)$. If the selected carrier anchor is the $\tau_2 = 0$ fiber point, so that

$$\alpha_{2,*} = \alpha_{2,m_Z}, \quad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

then the coherent validation couplings are

$$\alpha_{2,\dagger} = \alpha_{2,*} + \delta\alpha_2^\dagger, \quad \alpha_{Y,\dagger} = \alpha_{Y,*} + \delta\alpha_Y^\parallel + \delta\alpha_Y^\perp,$$

and they emit one coherent quintet

$$M_W^\dagger = v\sqrt{\pi\alpha_{2,\dagger}}, \quad M_Z^\dagger = v\sqrt{\pi(\alpha_{Y,\dagger} + \alpha_{2,\dagger})},$$

$$\alpha_{\text{em},\dagger}^{-1} = \frac{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}{\alpha_{Y,\dagger}\alpha_{2,\dagger}}, \quad \sin^2\theta_{W,\dagger} = \frac{\alpha_{Y,\dagger}}{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}.$$

The frozen-target electroweak validation law is one unique coherent value-emission law on top of the closed fiber map.

For the local PDG-2025 comparison surface,

$$\begin{aligned} \alpha_{2,m_Z} &= 0.03377843630219015, \\ \alpha_{Y,m_Z} &= 0.010131601067241624, \\ \eta_{\text{source}} &= 0.022147000871961295, v = 246.76711732749683 \text{ GeV}, \\ M_W^\dagger &= 80.377 \text{ GeV}, \\ M_Z^\dagger &= 91.18797809193725 \text{ GeV}, \end{aligned}$$

and the law emits

$$\begin{aligned} \tau_{2,W}^\dagger &= -2.3116268316325517 \times 10^{-4}, \\ \delta\alpha_2^\dagger &= -7.808313968675484 \times 10^{-6}, \\ M_{Z,\text{fiber}}^\dagger &= 91.17717018044121 \text{ GeV}, \\ \delta M_Z^\perp &= 10.807911496044653 \text{ MeV}, \\ \delta\alpha_Y^\parallel &= 2.3421435088707883 \times 10^{-6}, \\ \delta\alpha_Y^\perp &= 1.0302892666790446 \times 10^{-5}, \\ \delta n^\dagger &= 2.3463639031113336 \times 10^{-4}. \end{aligned}$$

So the coherent validation couplings are

$$\begin{aligned} \alpha_{2,\dagger} &= 0.03377062798822147, \\ \alpha_{Y,\dagger} &= 0.00969547694807616, \end{aligned}$$

and the same coherent branch emits

$$\begin{aligned} M_W^\dagger &= 80.377 \text{ GeV}, \quad M_Z^\dagger = 91.18797809193725 \text{ GeV}, \\ \alpha_{\text{em},\dagger}^{-1} &= 132.752408550967, \\ \sin^2\theta_{W,\dagger} &= 0.223058333896849. \end{aligned}$$

These two carrier-side scalars are bookkeeping data on the frozen validation surface. They are not the public electromagnetic readout on the Ward-projected D10 lane. That freeze-once law provides an exact coherent validation surface. On that frozen authoritative validation surface the canonical W/Z pair is hit exactly, but only as compare-only validation. The intended public electroweak prediction surface is the target-free source-only transport law. Its public validation records are the source spectral measure payload, same-scheme remainder, and interval certificate. The displayed W/Z numbers below remain frozen-adaptor comparison values.

Theorem 6.2 (Target-free source-only electroweak mass transport law). *Let the D10 source-only emitted basis be*

$$Q_{\text{src}}(P) = (\alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\text{source}}(P), v(P)),$$

and define

$$\rho_{\text{EW}} := \frac{\alpha_{2,m_Z} - \alpha_{Y,m_Z}}{\alpha_{2,m_Z} + \alpha_{Y,m_Z}}, \quad \alpha_u^{\text{seed}} := \frac{\eta_{\text{source}}}{\rho_{\text{EW}}}, \quad \lambda_{\text{EW}} := \frac{\eta_{\text{source}} \alpha_u^{\text{seed}}}{4} = \frac{\eta_{\text{source}}^2}{4\rho_{\text{EW}}}.$$

Then the beyond-selected-carrier transport map is uniquely

$$\begin{aligned} \tau_2^{\text{exact}} &= -\lambda_{\text{EW}} \left(1 + \frac{2}{3} \eta_{\text{source}} + \left(1 - \frac{\rho_{\text{EW}}}{6} \right) \eta_{\text{source}}^2 \right), \\ \delta n^{\text{exact}} &= \lambda_{\text{EW}} \left(1 + \frac{4}{3} \eta_{\text{source}} + \left(2 - \frac{\rho_{\text{EW}}}{6} \right) \eta_{\text{source}}^2 \right), \end{aligned}$$

with fiber hypercharge transport

$$\tau_Y^{\text{fiber}} = -\frac{\tau_2^{\text{exact}} + 2\eta_{\text{source}}}{1 + 4(\tau_2^{\text{exact}})^2}.$$

The coherent transport shifts are

$$\begin{aligned} \delta \alpha_2 &= \alpha_{2,m_Z} \tau_2^{\text{exact}}, \quad \delta \alpha_Y^{\parallel} = \alpha_{Y,m_Z} \frac{8\eta_{\text{source}}(\tau_2^{\text{exact}})^2 - \tau_2^{\text{exact}}}{1 + 4(\tau_2^{\text{exact}})^2}, \\ \delta \alpha_Y^{\perp} &= (\alpha_{2,m_Z} + \alpha_{Y,m_Z}) \delta n^{\text{exact}}. \end{aligned}$$

With

$$\alpha_{2,*} = \alpha_{2,m_Z}, \quad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

the coherent transport couplings are

$$\alpha'_2 = \alpha_{2,*} + \delta \alpha_2, \quad \alpha'_Y = \alpha_{Y,*} + \delta \alpha_Y^{\parallel} + \delta \alpha_Y^{\perp},$$

and they emit one coherent family

$$\begin{aligned} W &= v \sqrt{\pi \alpha'_2}, \quad Z = v \sqrt{\pi(\alpha'_2 + \alpha'_Y)}, \quad v = v(P). \\ a_0(P) &:= \frac{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P) \alpha_{Y,m_Z}(P)}, \quad s_0(P) := \frac{\alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}. \end{aligned}$$

So the target-free source-only transport law fixes the public mass pair together with the source-locked running-family anchor (a_0, s_0, v) from the D10 basis alone. The physical electromagnetic row is read from the Ward-projected unbroken $U(1)_Q$ channel in Theorem 6.3.

Using the same frozen electroweak validation basis

$$\begin{aligned} \alpha_{2,m_Z} &= 0.03377843630219015, \\ \alpha_{Y,m_Z} &= 0.010131601067241624, \\ \eta_{\text{source}} &= 0.022147000871961295, \\ v &= 246.76711732749683 \text{ GeV}, \end{aligned}$$

the target-free transport law would emit

$$\begin{aligned} W &= 80.37700001539531 \text{ GeV}, \quad Z = 91.18797807794321 \text{ GeV}, \\ a_0 &= 128.30576920234813, \\ s_0 &= 0.23073542347506173. \end{aligned}$$

Theorem 6.3 (Ward-projected $U(1)_Q$ transport law and Thomson-limit electromagnetic read-out). *Let the D10 source-only running family be locked by the forward pixel solve on the declared running/matching/threshold/scheme surface, with source basis*

$$Q_{\text{src}}(P) = (\alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\text{source}}(P), v(P)).$$

Assume the realized D9 branch, so that

$$Q = T_3 + Y$$

on the realized low-energy branch and color-singlet physical states carry integer Q -charge. Assume further that the post-selector electroweak transport object is the populated quotient-local kernel

$$K_{D10}^{\text{EW}}(q^2; P) = \{\Pi_{AA}(q^2; P), \Pi_{AZ}(q^2; P), \Pi_{ZZ}(q^2; P), \Pi_{WW}(q^2; P)\}$$

on the declared physical observable algebra, together with a Ward projector

$$\mathcal{W}_Q : K_{D10}^{\text{EW}}(q^2; P) \longrightarrow \Pi_Q(q^2; P)$$

onto the unbroken electromagnetic channel satisfying at $q^2 = 0$

$$\mathcal{W}_Q[\Pi_{AZ}(0; P)] = 0, \quad \mathcal{W}_Q[\Pi_{AA}(0; P)] \neq 0.$$

Assume the abelian projected edge-sector probabilities obey the $U(1)$ heat-kernel law

$$p_n(q^2; P) \propto e^{-t_Q(q^2; P)\lambda_n}, \quad \lambda_n = n^2,$$

equivalently

$$g_Q^2(q^2; P) = \frac{t_Q(q^2; P)}{2\pi},$$

and that the scalar readout package satisfies the exact provenance lock

$$\begin{aligned} \text{family_source_id} &= \text{d10_running_tree}, \\ \text{scheme_id} &= \text{freeze_once}, \\ \text{origin_kernel_id} &= \text{EWTransportKernel_D10}. \end{aligned}$$

Then the unique electromagnetic coupling readout on the Ward-projected D10 lane is

$$\alpha_{\text{em}}^{-1}(q^2; P) = \frac{8\pi^2}{t_Q(q^2; P)}.$$

If

$$a_0(P) := \alpha_{\text{em}}^{-1}(m_Z^2; P),$$

then the unique zero-normalized affine scalar on the same source-locked family is

$$\delta_\alpha(q^2; m_Z^2; P) := \frac{t_Q(m_Z^2; P)}{t_Q(q^2; P)} - 1,$$

so that

$$\alpha_{\text{em}}^{-1}(q^2; P) = a_0(P)(1 + \delta_\alpha(q^2; m_Z^2; P)).$$

Hence the Thomson-limit electromagnetic readout on that same Ward-projected electromagnetic kernel:

$$\alpha_{\text{Th}}^{-1}(P) := \lim_{q^2 \rightarrow 0} \alpha_{\text{em}}^{-1}(q^2; P) = a_0(P) \frac{t_Q(m_Z^2; P)}{t_Q(0; P)}.$$

Consequently, the low-energy electromagnetic row on this lane is read as the Thomson endpoint of the D10 electromagnetic transport family instead of as a mass-chart byproduct.

Proof. The realized D9 branch fixes the electric charge operator by $Q = T_3 + Y$, and the quotient-protected charge theorem fixes the physical color-singlet Q -lattice to be integral. The abelian edge-sector theorem supplies the heat-kernel extraction rule

$$p_n \propto e^{-t_Q \lambda_n}, \quad g_Q^2 = \frac{t_Q}{2\pi},$$

so the Ward-projected Q -channel determines a unique transport time $t_Q(q^2; P)$. Because the Ward projector kills $A - Z$ mixing at $q^2 = 0$, the selected channel is the physical unbroken electromagnetic channel. Then

$$\alpha_{\text{em}} = \frac{g_Q^2}{4\pi} = \frac{t_Q}{8\pi^2},$$

hence

$$\alpha_{\text{em}}^{-1} = \frac{8\pi^2}{t_Q}.$$

Evaluating at $q^2 = m_Z^2$ defines the source-locked anchor $a_0(P)$, and dividing by the same expression at general q^2 yields

$$\delta_\alpha = \frac{t_Q(m_Z^2)}{t_Q(q^2)} - 1.$$

The provenance-equality clause forces one running-family source, one frozen scheme, and one origin kernel for all scalar readouts, so no mixed-family or Z -only surrogate is promotable. The Thomson formula is the $q^2 \rightarrow 0$ limit of the same identity. $\square$

Corollary 6.4 (Ward-projected Maxwell normalization). *On the same Ward-projected $U(1)_Q$ lane, the electromagnetic field strength and current obey*

$$F_Q = dA_Q, \quad dF_Q = 0, \quad d*F_Q = g_Q^2(q^2; P)*J_Q,$$

with

$$g_Q^2(q^2; P) = \frac{t_Q(q^2; P)}{2\pi}, \quad \alpha_{\text{em}}^{-1}(q^2; P) = \frac{8\pi^2}{t_Q(q^2; P)}.$$

At the Thomson endpoint, this is the Maxwell normalization used by the public fine-structure row.

Proof. The compact reconstruction supplies the ordinary photon branch as the unbroken abelian $U(1)_Q$ factor with charge generator $Q = T_3 + Y$. On an abelian factor the compact-gauge curvature restricts to $F_Q = dA_Q$, so $d^2 = 0$ gives $dF_Q = 0$. Varying the quadratic electromagnetic action in the OPH coupling convention gives $d*F_Q = g_Q^2*J_Q$. The theorem above identifies the same branch coupling as $g_Q^2 = t_Q/(2\pi)$, hence $\alpha_{\text{em}}^{-1} = 8\pi^2/t_Q$. $\square$

Corollary 6.5 (Source-only release criterion on the Ward-projected D10 lane). *The Ward-projected D10 lane supports a source-only fine-structure release row only when the populated Ward-projected transport kernel satisfies*

$$\lim_{q^2 \rightarrow 0} \alpha_{\text{em}}^{-1}(q^2; P) = \alpha_{\text{Th}}^{-1}(P),$$

on the same source-locked family and without any separate endpoint value feeding that lane. Concretely, the release packet must contain `WARD_Q_SPECTRAL_MEASURE_RECEIPT`, `J24Q_SOURCE_JACOBI_RECEIPT`, `OMEGAQ_NORMALIZATION_RECEIPT`, `XI_SAME_SCHEME_REMAINDER_RECEIPT`, `NO_THOMSON_TARGET_LEAK_DAG`, `FULL_PIXEL_CONTRACTION_INTERVAL`, and `ALPHA_PREDICTION_BEATS_DIRECT_ME` under the direct-measurement interval adopted for that release. Until those objects are emitted, the public endpoint remains an empirical hadron-closure row.

This release criterion is the two-current marginal of the larger OPH-QCD hadronic precision backend. It is adequate for the Thomson endpoint and HVP branch only if the same source law also records the QCD quotient ensemble, source parameter map, Ward current normalization, systematics ledger, and no-target-leak DAG. Full hadronic precision claims require the higher-point and transition spectral exports from that same backend rather than reusing the scalar fine-structure residual.

The endpoint table records the source-locked anchor from the runtime candidate,

$$a_0(P) = 128.308268045165213892552005990181778935450 \dots,$$

and the OPH plus empirical hadron closure display value

$$\alpha_{\text{Th}}^{-1} = 137.035999177.$$

The repository's `code/P_derivation` witness path records the source-side audit trunk without a built-in reference inverse- α value. The public endpoint value is excluded as a source-only closure-solve input. The source-side audit trunk emits

$$\alpha_{\text{root}}^{-1} = 136.994835164621649457949994585787193262029 \quad ,$$

so the difference from the endpoint readout is recorded as an endpoint-and-matching audit packet.

Remark 6.6. *The empirical closure reference is the Thomson-limit inverse fine-structure constant. The source anchor $a_0(P) = 128.308268045165213892552005990181778935450 \dots$ is the electroweak-scale running-family value at m_Z^2 , while the display row is the low-energy endpoint. The 2022 CODATA/NIST reference value is $\alpha^{-1} = 137.035999177(21)$ [6].*

The freeze-once coherent repair surface provides compare-only validation, with machine-scale deltas

$$\Delta M_W \approx 1.54 \times 10^{-8} \text{ GeV}, \quad \Delta M_Z \approx -1.40 \times 10^{-8} \text{ GeV}.$$

The companion carrier-side pair

$$\begin{aligned} \alpha_{\text{em},\dagger}^{-1} &= 132.752408550967, \\ \sin^2 \theta_{W,\dagger} &= 0.223058333896849 \end{aligned}$$

is mass-chart bookkeeping on that frozen validation surface. It is not the public electromagnetic readout on the Ward-projected $U(1)_Q$ lane.

The electroweak quantitative-closure branch also contains several subordinate theorem objects: an underdetermination theorem for the quadratic repair family, the smallest theorem route through `ColorBalancedQuadraticRepairDescentEW`, the source-only mass emitter beneath the public W/Z rows, and the Ward-projected electromagnetic transport theorem above the source-locked running-family anchor. The scalar-package consistency clause is the shared provenance lock

$$(\text{d10_running_tree}, \text{freeze_once}, \text{EWTransportKernel_D10}).$$

A further forward transmutation certificate sits directly beneath the promoted readout and records that the source-only basis reconstructs the same $\alpha_U(P)$, $t_U(P)$, and $t_{\text{tr}}(P)$ as the pixel-closure solve without reading them back from measured couplings. These objects locate the public repair law inside the quantitative-closure branch without changing the emitted public surface.

6.2 Hierarchy interpretation on the local transmutation branch

The selected exact OPH branch solves the electroweak hierarchy/naturality problem. The large cosmic horizon ratio and the electroweak hierarchy belong to different OPH branches. The global capacity branch N_{CRC} accounts for the cosmic area and horizon-length hierarchy. The weak scale is a local pixel/transmutation readout:

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P_\star)}\right].$$

The hierarchy certificate records the interval

$$I_U = [0.041123336195630494, 0.041125336195630496],$$

the Krawczyk inclusion

$$K(I_U) \subset [0.0411243357185544983, 0.0411243366727064662] \subset \text{int}(I_U),$$

and a strictly negative derivative enclosure $[-10.995768, -10.985284]$. Thus the local source equation has a unique zero in that enclosure. On the public endpoint branch,

$$\begin{aligned} \alpha_U(P_\star) &= 0.041124336195630495, \\ \frac{v}{E_\star} &= 2.0199803239725553 \times 10^{-17}. \end{aligned}$$

The source-audit branch gives

$$\begin{aligned} P_{\text{cand}} &= 1.63097209569432901817967892561191884270169, \\ \frac{v(P_{\text{cand}})}{E_\star} &= 2.0198114150099223 \times 10^{-17}, \end{aligned}$$

with the public Thomson endpoint excluded upstream. The Planck-to-weak ratio is generated by the ordinary exponential transmutation factor controlled by the unified diffusion coupling. The twenty-four-round capacity contraction belongs to the global horizon-length ratio.

This is also the OPH reading of Higgs naturalness on the declared D10/D11 surface. The observer-visible scalar mass is the normal-form readout

$$m_H = H_{\text{OPH}}(P_\star)$$

on the selected branch. The split into a bare Higgs mass plus a cutoff correction is a regulator coordinate split outside the observer-facing physical variable. A scalar relevant deformation absent from the retained branch constraints is off-branch; a scalar deformation present on the selected branch has its value fixed by the branch equations. The claim inherits the D10/D11 quantitative surface and the $\mathcal{R}_U$ precision policy. The selected exact source-to-Higgs coarse-graining square has $\epsilon_H = 0$, with $\epsilon_H \in [0, 0]$, as stated below.

6.3 Local/global resonance continuation for the hierarchy

The local/global hierarchy resonance compares the D10 transmutation exponent with the global capacity branch at the same joint pair $(P_\star, N_{\text{CRC}})$. The target relation is

$$t_{\text{tr}}(P_\star) = \frac{P_\star}{12} \log\left(\frac{N_{\text{CRC}}}{\pi}\right),$$

or equivalently

$$\frac{v}{E_{\text{cell}}} = \left(\frac{N_{\text{CRC}}}{\pi} \right)^{-P_{\star}/12}.$$

Writing

$$|g'_{\star}| = \left(\frac{N_{\text{CRC}}}{\pi} \right)^{-1/48}$$

turns the same expression into

$$\frac{v}{E_{\text{cell}}} = |g'_{\star}|^{4P_{\star}}.$$

This packages the electroweak hierarchy as a resonance between the local pixel fixed point and the global screen-capacity fixed point. The screen branch supplies twelve irreducible curvature ports. Including reversible write/check orientation gives a 24-slot oriented repair register, and the hierarchy exponent uses this 24-fold repair normalization through $m_{\text{rep}} = 24$. The local weak hierarchy reads the global screen depth through a $4P_{\star}$ electroweak projection.

The repair count is fixed by the observer-visible product-adjoint spectrum. On the realized product branch,

$$m_{\text{rep}} = 2 \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8 + 3 + 1) = 24.$$

The factor 2 is the reversible write/verify orientation of each repair channel. The $\mathfrak{su}(5)$ adjoint has the same single-orientation integer for a different support: its X/Y mixed gauge channels are absent from the OPH product branch.

Lemma 6.7 (Global repair-tick lemma). *Let $N_{\text{CRC}} = F(N_{\text{CRC}})$ be the cosmic record-capacity fixed point with observed-branch readout $N_{\text{CRC}} = S_{\text{dS}}$ and D6 normalization*

$$N_{\text{CRC}} = \pi \left(\frac{r_{\text{CRC}}}{\ell_{\star}} \right)^2.$$

Work in the multiplicative screen-radius coordinate $\rho = \ell/r_{\text{CRC}}$, so the horizon sits at $\rho = 1$ and the local cell at $\rho_{\star} = \ell_{\star}/r_{\text{CRC}}$. Define the global repair cycle G_N at the fixed point as the scale-free readback transport of the horizon record coordinate, with fixed-point closure (readback without deficit or slack) meaning $G_N(1) = \rho_{\star}$ exactly. The representation-to-spectrum theorem gives $m_{\text{rep}} = 24$, so the selected resonance branch decomposes as $G_N = g_N^{24}$ with g_N the positive root. Then

$$G_N(\rho) = \left(\frac{N_{\text{CRC}}}{\pi} \right)^{-1/2} \rho, \quad g_N(\rho) = \left(\frac{N_{\text{CRC}}}{\pi} \right)^{-1/48} \rho, \quad |g'_{\star}| = \left(\frac{N_{\text{CRC}}}{\pi} \right)^{-1/48}.$$

More generally an m -tick decomposition gives $|g'_{\star}| = (N_{\text{CRC}}/\pi)^{-1/(2m)}$; $m_{\text{rep}} = 24$ gives the stated exponent.

Proof. The D6 normalization gives $\rho_{\star} = \ell_{\star}/r_{\text{CRC}} = (N_{\text{CRC}}/\pi)^{-1/2}$. The closure condition follows at the readback interface. The Observers synthesis defines the readback map as $F(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}_{r,N}(\mathfrak{U}_{r,N})))$ in the refinement limit, the capacity reconstructed by the interior observer sector from the horizon record. Modeling Cap_{read} by the D6 area law at the effective delivery resolution ρ_{read} , the canonical scale-covariant extension of $N = \pi(r/\ell)^2$, gives the D6-consistent counting identification. The corpus defines Cap_{read} as reconstructed capacity; this certificate adds the D6-consistent counting rule. The readback reconstructs $F(N) = \pi/\rho_{\text{read}}^2$, so the fixed-point equation $N_{\text{CRC}} = F(N_{\text{CRC}}) = \pi/\rho_{\star}^2$ forces $\rho_{\text{read}} = \rho_{\star}$ for positive roots: deficit ($\rho_{\text{read}} > \rho_{\star}$) gives $F(N) < N$, slack ($\rho_{\text{read}} < \rho_{\star}$) gives $F(N) > N$. This is exactly $G_N(1) = \rho_{\star}$. Homogeneity forces $G_N(\rho) = Q\rho$ with $Q > 0$, hence $Q = (N_{\text{CRC}}/\pi)^{-1/2}$. The homogeneous one-tick map with $g_N^{24} = G_N$ is the positive 24th root, so its multiplier, and hence its absolute derivative in this coordinate, is $(N_{\text{CRC}}/\pi)^{-1/(2 \cdot 24)} = (N_{\text{CRC}}/\pi)^{-1/48}$. $\square$

The emitted particle rows continue to use the $\mathcal{R}_U$ certificate above. Lemma 6.7 closes the global tick side with the derived representation-to-spectrum count $m_{\text{rep}} = 24$. The proof derives the closure transport at the interface level of F under the D6 area-law counting model of Cap_{read} .

Theorem 6.8 (Joint product fixed-point branch). *Let $x = \log N$, let $X = I_P \times I_x$, and equip X with*

$$d((P, x), (Q, y)) = \max\{|P - Q|, |x - y|\}.$$

On the product-separated source branch the joint map is

$$\mathcal{J}(P, x) = (\Gamma(P), \hat{C}(x)), \quad \hat{C}(x) = \log F(e^x),$$

where Γ is the local pixel closure map and F is the global capacity readback map. If

$$|\Gamma(P) - \Gamma(Q)| \leq q_P |P - Q|, \quad |\hat{C}(x) - \hat{C}(y)| \leq q_N |x - y|, \quad q_P, q_N < 1,$$

then $\mathcal{J}$ has a unique fixed point $(P_\star, \log N_{\text{CRC}})$, and every iteration in X converges to it with contraction constant $q = \max\{q_P, q_N\}$.

Proof. For any two points in X ,

$$d(\mathcal{J}(P, x), \mathcal{J}(Q, y)) \leq \max\{q_P |P - Q|, q_N |x - y|\} \leq \max\{q_P, q_N\} d((P, x), (Q, y)).$$

Banach's theorem gives existence, uniqueness, and stability. A genuinely coupled source map $\mathcal{J}(P, x) = (\Gamma(P, x), \hat{C}(P, x))$ requires derivative bounds a, b, c, d and a weight $r > 0$ with

$$\max\{a + b/r, d + rc\} < 1$$

in the weighted sup metric. Without that bound, the coupled branch has residual freedom and no uniqueness theorem. $\square$

Theorem 6.9 (Electroweak tick-projection bridge). *On the selected D10 source branch,*

$$\frac{v}{E_{\text{cell}}} = \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

Together with the global repair tick

$$|g'_\star| = \left(\frac{N}{\pi}\right)^{-1/48},$$

define the electroweak projection exponent by

$$\Pi_{\text{EW}}(P, N) := \frac{-\log(v/E_{\text{cell}})}{-\log |g'_\star|} = \frac{24\pi}{\alpha_U(P) \log(N/\pi)}.$$

Since $0 < |g'_\star| < 1$, this exponent is unique. The target projection

$$\Pi_{\text{EW}}(P_\star, N_{\text{CRC}}) = 4P_\star$$

is equivalent to the bridge residual

$$\mathcal{B}_{\text{EW}}(P, N) := \alpha_U(P) \log(N/\pi) - \frac{6\pi}{P} = 0.$$

Equivalently, the exact bridge target is

$$N_{\text{EW}}(P) = \pi \exp \left[\frac{6\pi}{P\alpha_U(P)} \right].$$

For the public endpoint branch,

$$N_{\text{EW}}(P_\star) = 3.5323546226929906511187512962330547600462 \times 10^{122}.$$

For the bridge-refined capacity certificate

$$N_{\text{CRC}}^{\text{EW}} = \pi \exp \left[\frac{6\pi}{P_\star \alpha_U(P_\star)} \right],$$

one has $\mathcal{B}_{\text{EW}}(P_\star, N_{\text{CRC}}^{\text{EW}}) = 0$, and hence

$$\frac{v}{E_{\text{cell}}} = \left(\frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_\star/12} = |g'_\star|^{4P_\star}.$$

The factor 4 is the electroweak transmutation multiplicity $\beta_{\text{EW}} = N_c + 1 = 4$. The factor 48 is $2 \cdot 24$ from the global repair tick, hence $12 = 48/4$. The rounded $N_{\text{CRC}} \simeq 3.31 \times 10^{122}$ display is a capacity-scale label and fails the exact bridge residual.

Proof. The D10 relation gives

$$-\log(v/E_{\text{cell}}) = \frac{2\pi}{4\alpha_U(P)} = \frac{\pi}{2\alpha_U(P)}.$$

The global tick relation gives

$$-\log |g'_\star| = \frac{1}{48} \log(N/\pi).$$

Taking the ratio gives the displayed Π_{EW} . Setting $\Pi_{\text{EW}} = 4P$ and rearranging gives $\alpha_U(P) \log(N/\pi) = 6\pi/P$, exactly $\mathcal{B}_{\text{EW}}(P, N) = 0$. Substitution gives the displayed $N_{\text{EW}}(P)$ and the hierarchy identity. $\square$

Theorem 6.10 (Icosahedral screen sieve and capacity projection). *On the OPH triangulated S^2 screen branch with a locally six-valent MaxEnt quotient-ensemble background, convex defect cost, edge-center collar readout, and no-marked-point finite isotropy, the screen has exactly twelve irreducible curvature ports. The ports form the 12-vertex icosahedral orbit, and an invariant screen load X is read locally as $X/12$. For $X = \log(N/\pi)$, local cell entropy $P/4$, and electroweak multiplicity $\beta_{\text{EW}} = 4$,*

$$\Gamma_{\text{EW}} = \beta_{\text{EW}} \frac{P}{4} \frac{1}{12} \log(N/\pi) = \frac{P}{12} \log(N/\pi).$$

Consequently

$$\log(E_{\text{cell}}/v) = \frac{P}{12} \log(N/\pi), \quad \alpha_U(P)^{-1} = \frac{P}{6\pi} \log(N/\pi),$$

which is equivalent to the electroweak bridge residual $\mathcal{B}_{\text{EW}}(P, N) = 0$.

Proof. For a triangulated sphere with V vertices, E edges, and F triangular faces, $V - E + F = 2$, $3F = 2E$, and hence $E = 3V - 6$. With defect charge $q_v = 6 - \deg(v)$,

$$\sum_v q_v = 6V - 2E = 12.$$

The local vacuum is six-valent, so the total spherical curvature appears as positive defects. Convex defect cost spreads the fixed charge 12 into twelve unit $q_v = 1$ defects. Tetrahedral and octahedral alternatives carry the same total charge while concentrating it into four $q = 3$ defects or six $q = 2$ defects, placing them outside the MaxEnt/refinement-stable screen branch. Edge-center collars expose each unit fivefold defect as one central port, and no-marked-point finite isotropy places the twelve ports on the maximal finite rotation orbit of S^2 . For $I \simeq A_5$, orbit-stabilizer gives $|I|/|C_5| = 60/5 = 12$. An invariant scalar contraction on that transitive orbit is split equally, giving the $1/12$ sieve factor. Multiplying the sieved screen load by $P/4$ and $\beta_{EW} = 4$ gives $\Gamma_{EW} = (P/12) \log(N/\pi)$. Exponentiating gives the displayed hierarchy relation, and matching it to the source transmutation exponent gives the displayed α_U equation and $\mathcal{B}_{EW}(P, N) = 0$. $\square$

Theorem 6.11 (EW-refined exact capacity fixed point). *Define the log-capacity map*

$$\mathcal{C}_{EW}(P, x) = (1 - \lambda)x + \lambda \frac{6\pi}{P\alpha_U(P)}, \quad 0 < \lambda \leq 1.$$

For fixed $P = P_\star$ and $\lambda = 1/2$, this is a contraction with Lipschitz constant $1/2$. Its unique fixed point is

$$x_\star = \frac{6\pi}{P_\star \alpha_U(P_\star)}, \quad N_{CRC}^{EW} = \pi e^{x_\star}.$$

Therefore

$$\mathcal{B}_{EW}(P_\star, N_{CRC}^{EW}) = 0.$$

The rounded 3.31×10^{122} capacity display remains diagnostic for this bridge.

Proof. The map has derivative $1 - \lambda$ in x , hence is a contraction for $0 < \lambda \leq 1$. Solving $x = \mathcal{C}_{EW}(P_\star, x)$ gives the displayed $x_\star$. Substitution into $\mathcal{B}_{EW}$ gives zero. $\square$

Theorem 6.12 (RG/Higgs naturality closure). *On the selected exact hierarchy branch, let Q_s be the source hierarchy quotient and Q_H the Higgs/electroweak quotient. Define*

$$\rho_{sH}([x]_s) = [P(x), N(x), \Theta(P(x), N(x)), \Pi_{HT} F_{D11} F_{D10}(P(x), N(x)), 0]_H,$$

with

$$\Theta(P, N) = \left(\frac{N}{\pi}\right)^{-P/12} = |g'_\star|^{4P}.$$

For the corresponding normal-form maps and obstruction maps,

$$\rho_{sH} n_s = n_H \rho_{sH}, \quad \chi_{sH} h_s = h_H \rho_{sH}.$$

Consequently,

$$\varepsilon_H = \max\{\varepsilon_{sH}^n, \varepsilon_{sH}^h\} = 0, \quad \varepsilon_H \in [0, 0].$$

Proof. The two routes through the source and Higgs quotients preserve the same tuple

$$(P, N, \Theta(P, N), \Pi_{HT} F_{D11} F_{D10}(P, N)).$$

The equality with $|g'_\star|^{4P}$ follows from $|g'_\star| = (N/\pi)^{-1/48}$. A hidden scalar relevant deformation is not an independent physical input on the selected branch: if absent from the retained finite constraint family, it is off-branch by refinement stability; if retained, its value is fixed by the branch equations and belongs to the displayed Higgs/top coordinate. Both controlled squares commute exactly, so the product defect is zero. $\square$

The RG/Higgs naturality closure does not use the measured weak scale. Measured W/Z , Higgs/top, G , Planck-area, and Λ data are excluded inputs. The local/global package records the closed global repair tick, joint product branch, electroweak projection bridge, EW-refined exact capacity fixed point, finite readback-resolution certificate, and RG naturality square. This is the OPH solution of the electroweak hierarchy/naturality problem on the selected exact branch. The finite readback certificate defines

$$F_r(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}_{r,N}(\mathfrak{U}_{r,N}))), \quad \rho_{\text{read}}(r, N) = \sqrt{\pi/F_r(N)},$$

and proves one selected effective delivery resolution on the finite branch, with $\rho_{\text{read}}(r, N_{\text{CRC}}) \rightarrow (N_{\text{CRC}}/\pi)^{-1/2}$ in the positive-root refinement limit. The representation-to-spectrum theorem gives $m_{\text{rep}} = 2(8 + 3 + 1) = 24$, the 24-slot oriented repair register on the selected branch, so the local/global hierarchy-resonance package closes on the selected branch. The promoted hierarchy row does not use the local/global resonance as an input.

6.4 Higgs/top critical stage

The Higgs/top critical stage is a quantitative-theorem stage built on top of the electroweak gauge core. It is described by a source-only split law together with a Jacobian readout map on the declared D10/D11 running, matching, and threshold surface. The promoted D10 tuple

$$(\eta_{\text{source}}, \beta_{EW}, \lambda_{EW}, \tau_{2,\text{tree}}^{\text{exact}}, \delta n_{\text{tree}}^{\text{exact}})$$

emits the shared scalar

$$\rho_{HT} = \log(1 + \tau_{2,\text{tree}}^{\text{exact}})$$

and the source-only residual selectors

$$R_T = -\tau_{2,\text{tree}}^{\text{exact}} \eta_{\text{source}}^2 + \left(1 + \frac{\beta_{EW}}{28}\right) \eta_{\text{source}}^6 + \frac{\eta_{\text{source}}^8}{14} + \frac{\eta_{\text{source}}^9}{27},$$

$$R_H = \eta_{\text{source}}^5 - \frac{3}{25} \eta_{\text{source}}^6 + \frac{\lambda_{EW} \eta_{\text{source}}^6}{18} + \frac{\eta_{\text{source}}^8}{2\beta_{EW}}.$$

The split coordinates are

$$\pi_y = \frac{\eta_{\text{source}} + \left(\frac{3}{2} + \frac{\beta_{EW}}{4}\right) \rho_{HT} + R_T}{\sqrt{\pi}}, \quad \pi_\lambda = \frac{\eta_{\text{source}} - \left(\frac{4}{3} - \frac{\beta_{EW}}{54}\right) \rho_{HT} + R_H}{\sqrt{\pi}},$$

so the declared D11 Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y y_t^{\text{core}}(\mu_t), \quad \delta \lambda(\mu_t) = -\frac{16}{9} \pi_\lambda \lambda^{\text{core}}(\mu_t).$$

This gives

$$m_H = 125.1995304097179 \text{ GeV}, \quad m_t^{D11} = 172.3523553288312 \text{ GeV}.$$

At the precision quoted by the PDG Higgs listing, the Higgs row lands on the 2025 Higgs average. The same surface emits a companion top coordinate. The selected-class quark support witness carries a target-anchored running-top audit row using the PDG 2025 cross-section entry; it is not a separate public source-only top prediction. The bridge to the auxiliary direct-top PDG row is closed

as a corpus-limited codomain no-go; the auxiliary row is compare-only. The one-scalar companion seed

$$\sigma_{D11,HT} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

is on disk only as the fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{D11,HT}$. Separately, the same D11 Jacobian admits a compare-only inverse slice that hits the canonical Higgs/top reference pair exactly. That exact sidecar is useful for bookkeeping and does not define the forward branch used for the public rows. The Higgs boson belongs naturally in the boson discussion, while the top quark will be revisited in the quark-family chapter as the third-generation up-type quark. The top coordinate reported here is the D11 split-theorem coordinate on this surface, not a separate public top-mass prediction row. The selected-class quark support wrapper carries only a target-anchored audit witness under the strict public-output policy.

The distinction is between the closed source-only split theorem and the broader paper surface. The split theorem is closed on its emitted D10/D11 branch, so the Higgs/top rows are substantive emitted outputs. The exact inverse slice is a compare-only validation surface, and the theorem is confined to the declared D10/D11 quantitative surface instead of promoted into the recovered core or into the selected-class quark support wrapper.

The electroweak chapter therefore presents three mathematically distinct surfaces: the target-free source-only theorem surface that would supply promoted prediction rows after the declared RG/matching/threshold/scheme surface is internally certified, the selected-carrier chart beneath it, and the freeze-once coherent repair surface used for the compare-only W/Z values. That is the technically direct way to present the bosonic sector while keeping validation bookkeeping and matching/scheme gates separate from the declared P -closure and fine-structure endpoint.

7 Flavor Transport, Generation Structure, and the Yukawa Dictionary

The gauge branch explains why the realized low-energy sector has three generations, but it does not by itself produce the full flavor dictionary. This paper therefore needs a separate flavor chapter. The imported realized-branch input here is the D9 count package $N_c = 3$ and $N_g = 3$. This chapter is the bridge between that structural result and the matter-family chapters for quarks, charged leptons, and neutrinos.

The flavor derivation is deliberately constructive. It does not claim that the full OPH flavor observable is theorem-level. Instead, it builds the object chain that a closed theorem would have to pass through. The derivation chain has the following main architecture: start from a refinement-indexed family transport kernel, derive a centered generation-bundle branch generator, lift that to same-label transport data, derive the induced overlap-edge transport cocycle, reduce the cocycle to a persistent flavor observable, and then push that observable into common sector-response objects for the downstream quark, charged-lepton, and neutrino derivations.

Mathematically, this derivation is where “why this family exists” becomes a concrete technical question. The realized Standard Model branch gives three generations structurally; the flavor derivation is where one tries to turn that three-generation existence statement into transport, splitting, suppression, phase, and excitation data. The relevant objects are the intermediate transport and spectral structures, not the final fermion masses themselves, that later mass readouts consume.

The active chain, in slightly compressed form, is:

1. normalize a refinement-indexed family transport kernel;

2. derive a centered generation-bundle branch generator on the realized three-generation charged bundle;
3. lift this to same-label edge-line transport and then to an overlap-edge cocycle with explicit defect and gap bookkeeping;
4. reduce those data to projectors, spectral gaps, pair suppressions, and cycle phases;
5. push the resulting family object into sector-response objects and then into suppression/phase tensors for the downstream matter derivations.

Three lane-specific claim boundaries are attached to this region of the derivation. The shared excitation dictionary is the common proof-facing base. Above that base, the charged lane carries exact centered readback, a closed common-shift no-go, the declared same-label q_e readback, a source-side determinant character for any fixed formal exponent vector

$$S_M = \sum_e M_e^{\text{ch}} \log q_e,$$

a determinant-line lift on theorem-grade physical charged data, and an algebraic mass readout from theorem-grade $A_{\text{ch}}(P)$. The theorem lane does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The charged-lepton theorem gap is the determinant trace-lift attachment $3\mu(r) = S_M(r)$ on the charged determinant channel. The charged determinant channel has a corpus-limited no-go boundary: the uncentered trace lift is not emitted. The neutrino lane carries a target-informed weighted-cycle candidate above a template family kernel, with compare-only bridge and absolute-attachment diagnostics. The quark lane carries the internalized mass bridge together with a selected-class audit/support theorem on the public physical quark frame class f_P . Its physical sigma datum is target-derived, so the running sextet is withheld from public prediction tables while the affine mean law, selected-frame support data, and exact forward Yukawas Y_u and Y_d remain recorded. A separate compare-only microphysical bridge records edge-statistics transport on a diagnostic surface. Together these boundaries mark where the flavor derivation leaves the common transport backbone and passes to lane-specific closure contracts.

8 Quark Family Derivation

The quark lane carries a selected-class audit/support closure on the public physical quark frame class f_P chosen by P . On that selected class the physical sigma datum used by the exact sextet is target-derived, so the numeric running-mass sextet is withheld from public prediction tables. The affine mean law and ordered three-point readout remain useful support data, and the exact forward construction emits explicit forward Yukawas Y_u and Y_d on the selected frame. The same target-anchored sextet is also realized by two supporting exact surfaces: the exact same-family witness on `current_family_only` and the restricted common-refinement transport-frame theorem on `current_family_common_refinement_transport_frame_only`. This is selected-class closure only. It does not classify all public quark frame classes.

8.1 What quarks are in this derivation

Quarks are the color-charged elementary constituents of hadronic matter. In nature, the up quark and down quark dominate protons and neutrons, while the strange quark, charm quark, bottom

quark, and top quark appear in progressively heavier and more unstable sectors. In the OPH particle derivation, the quark lane has a closed downstream algebraic readout conditional on a source-only physical sigma datum. It also records explicit exact forward Yukawas on the selected public class f_P . That is a conditional support surface, not a public source-only mass prediction, because the current exact sigma datum is target-derived.

The quark route starts from the shared flavor excitation dictionary, internalizes the target-free mass bridge on the D12 mass ray, proves selected-bridge-fiber representative independence for the attached physical sigma datum on f_P , applies the affine mean law, and closes both the ordered three-point readout and the exact forward construction once a physical sigma datum is supplied. The same-label left-handed selector surface closes to the singleton σ_{ref} . That lower object is a negative sheet-selector statement on the selected D12 sheet. It does not replace the missing source-only sigma selector.

8.2 Emitted quark rows

These comparisons are made against running quark references, not direct free-particle pole masses, because quarks are confined. That is not a technicality; it is part of what the quark chapter must explain to technical readers. The current exact sextet matches the official PDG 2025 API running-quark target surface, with the top coordinate taken from the PDG cross-section entry, because the sigma datum has target ancestry. The auxiliary direct-top row is a compare-only extraction codomain with a corpus-limited no-go boundary. The Higgs/top branch also carries the same exact top value 172.3523553288312 GeV on the declared D11 split theorem surface. The table above records the quark-family conditional support surface on f_P ; it is withheld from public prediction columns until a source-only sigma selector closes.

8.3 Target-free mass bridge and selected-class public theorem

The same-label left-handed solver surface closes to the singleton σ_{ref} (`sigma_ref`). This is a negative same-sheet selector statement: same-sheet rephasing preserves CKM moduli, so it does not move that selected sheet to the physical CKM shell. Same-sheet overlap scans, chirality-swapped basis diagnostics, and other non-sector-attached orbit improvements are compare-only and do not change the selected-class theorem.

A separate target-free mass bridge is internalized on the emitted D12 ray. On the minimal light branch

$$y_u = c_u \varepsilon^6, \quad y_d = c_d \varepsilon^6, \quad \varepsilon = \frac{1}{6},$$

the light-quark overlap-defect theorem emits

$$\Delta_{ud}^{\text{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}.$$

The emitted same-family D12 mass object is the ray

$$D12_{ud}^{\text{mass}} \equiv \mathcal{R}_{D12}^{ud} \quad (\text{paper id D12_ud_mass_ray}),$$

and on that ray the same scalar is equivalently the one-scalar law

$$\Theta_{ud}^{\text{mass}} := \text{quark_same_family_value_law}.$$

The exact scalar identities are

$$\Delta_{ud}^{\text{overlap}} = \frac{t_1}{5}, \quad \log \frac{c_d}{c_u} = \frac{6}{5} t_1, \quad t_1 = 5 \Delta_{ud}^{\text{overlap}} = \frac{5}{6} \log \frac{c_d}{c_u}.$$

The D12 mass-side package is therefore functorial:

$$\begin{aligned}\eta_Q^{\text{centered}} &= -\frac{1-x_2^2}{27}t_1, \\ \kappa_Q &= -\frac{t_1}{54}, \\ x_2 &= -0.5175863354681689.\end{aligned}$$

The odd source package is also forced:

$$\begin{aligned}\beta_{u,\text{diag},B}^{\text{source}} &= \frac{t_1}{10}, \\ \beta_{d,\text{diag},B}^{\text{source}} &= -\frac{t_1}{10}, \\ \text{source_readback_u_log_per_side} &= \left(-\frac{t_1}{10}, 0, +\frac{t_1}{10}\right), \\ \text{source_readback_d_log_per_side} &= \left(+\frac{t_1}{10}, 0, -\frac{t_1}{10}\right),\end{aligned}$$

and with the emitted spread pair

$$\sigma_u = 5.5905, \quad \sigma_d = 3.3049,$$

one gets

$$\tau_u = \frac{\sigma_d}{10(\sigma_u + \sigma_d)}t_1, \quad \tau_d = \frac{\sigma_u}{10(\sigma_u + \sigma_d)}t_1.$$

The mass bridge is not part of the selected-class theorem boundary. The builder-facing pure- B payload pair is a lower implementation object beneath the exact theorem surface, and `intrinsic_scale_law_D12` is the derived wrapper above $\Theta_{ud}^{\text{mass}}$.

Write the same-label left-handed physical carrier as

$$\Sigma_{ud}^{\text{phys}} := \left\{ (\sigma_{\text{id}}, \tau, U_{u,L}, U_{d,L}, V_{\text{CKM}}, I_{\text{CKM}}) : V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L} \right\} / \sim,$$

where

$$(U_{u,L}, U_{d,L}, V) \sim (U_{u,L}D_u, U_{d,L}D_d, D_u^\dagger V D_d)$$

for diagonal $D_u, D_d \in U(1)^3$. On the selected public quark frame class f_P , represented on the realized lane by the explicit common-refinement transport-frame class $[F_0^\dagger F_1]$, the exact $\Sigma_{ud}^{\text{phys}}$ element and the attached exact sigma datum are independent of the declared representative inside the selected bridge fiber. Therefore the exact sigma readout descends uniquely to the selected audit/support surface on f_P . It is not a public source-only mass prediction because the physical sigma datum remains target-derived.

Sigma descent non-selection. Let $R_{\text{decl}}(f_P)$ be the declared selected bridge fiber over the public quark frame class. If a map

$$\Sigma : R_{\text{decl}}(f_P) \rightarrow \mathbb{R}^4$$

is constant on that fiber, then it descends to a well-defined public datum $\bar{\Sigma}(f_P)$. This proves representative independence only. It does not select the value of $\bar{\Sigma}(f_P)$, because every constant vector in $\mathbb{R}^4$ would also descend. Therefore a value obtained from the current-family running-quark target surface remains target-derived after descent.

Source sigma selector target. The missing promotion theorem is a source-only selector for

$$\Sigma_{ud}^{\text{phys}}(P) = (\sigma_u, \sigma_d, \sigma_{\text{seed}}^{ud}, \eta_{ud})$$

with no dependency path to PDG or current-family quark targets. The target value for a future source theorem is

$$\begin{aligned} \sigma_u &= 5.573928426395543, & \sigma_d &= 3.296264198808688, \\ \sigma_{\text{seed}}^{ud} &= 4.4350963126021155, & \eta_{ud} &= 1.1388321137934274. \end{aligned}$$

The required theorem package consists of a source sigma quotient, a source selector, refinement compatibility, and a no-target-leak DAG certificate.

Edge-statistics correction target. The strongest on-disk source-side candidate emits

$$\begin{aligned} \sigma_u^{\text{edge}} &= S_{13} + \frac{\rho_{\text{ord}} \delta_{21}}{1 + \rho_{\text{ord}}} = 5.578418804072826, \\ \sigma_d^{\text{edge}} &= S_{23} + \frac{\delta_{21}}{2(1 + \rho_{\text{ord}} - x_2^2)} = 3.4210589139721543. \end{aligned}$$

To close the source sigma theorem, OPH source objects must emit the correction terms

$$R_u = -0.004490377677282886, \quad R_d = -0.1247947151634663,$$

equivalently $R_{\text{seed}} = -0.06464254642037481$ and $R_\eta = 0.060152168743091705$, without defining them as residuals against the running-quark target surface.

Define

$$\begin{aligned} \sigma_{\text{seed}}^{ud} &:= \frac{\sigma_u + \sigma_d}{2}, & \eta_{ud} &:= \frac{\sigma_u - \sigma_d}{2}, \\ A_{ud} &:= \frac{1}{2(1 + \rho_{\text{ord}} - x_2^2)}, \\ B_{ud} &:= \frac{1}{2\left(1 - x_2^2 - \frac{x_2^2}{1 + \rho_{\text{ord}}}\right)}, \\ \rho_{\text{ord}} &= 1.294284936377706. \end{aligned}$$

Then

$$g_u = g_{\text{ch}} \exp(-(A_{ud} \sigma_{\text{seed}}^{ud} - B_{ud} \eta_{ud})), \quad g_d = g_{\text{ch}} \exp(-(A_{ud} \sigma_{\text{seed}}^{ud} + B_{ud} \eta_{ud})).$$

Given a source-only physical sigma datum, the affine mean law emits (g_u, g_d) algebraically on f_P . With the current target-derived sigma datum, the ordered three-point quadratic readout realizes the target-anchored running quark sextet as an audit witness, and the exact forward construction emits explicit exact forward Yukawas Y_u and Y_d on that selected public class. The conditional selected-class audit/support surface can be written as

$$f_P + \Sigma_{ud}^{\text{source}}(P) \implies (\sigma_u, \sigma_d, \sigma_{\text{seed}}^{ud}, \eta_{ud})_{\text{phys}} \implies (g_u, g_d) \implies (m_u, m_d, m_s, m_c, m_b, m_t),$$

together with explicit exact forward Yukawas Y_u and Y_d on the same selected-class surface. The first arrow $P \rightarrow \Sigma_{ud}^{\text{source}}(P)$ is the missing source theorem.

Maximal theorem-emitted quark package theorem. Let

$$\begin{aligned}
P &:= p_a \\
&+ (\text{Axioms 1--5}) \\
&+ \text{Assumption 6} \\
&+ \text{the emitted OPH nodes D1--D10} \\
&+ \text{the listed public D12 quark objects.}
\end{aligned}$$

Then the quark-side package can be written in four layers:

1. the emitted D12 mass ray

$$D12_{ud}^{\text{mass}} = \mathcal{R}_{D12}^{ud} = \left\{ \lambda \left(\frac{1}{5}, -\frac{1-x_2^2}{27} \right) : \lambda \geq 0 \right\}, \quad \lambda = \text{ray_modulus} = t_1;$$

2. the same-label left-handed selector value

$$\sigma_{ud} = \sigma_{\text{ref}},$$

with canonical token `D12::same_label_left::reference_sheet`, which is a negative sheet-selector statement;

3. the separate target-free mass bridge

$$P \vdash \Delta_{ud}^{\text{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}, \quad P \vdash \Theta_{ud}^{\text{mass}} = \text{quark_same_family_value_law};$$

4. the selected-class support wrapper on f_P , which proves representative independence for the attached physical sigma datum and conditionally carries the affine pair (g_u, g_d) , the running sextet, and explicit exact forward Yukawas Y_u and Y_d once a source-only sigma selector is supplied.

A separate exact same-family witness reproduces the same sextet

$$(u, d, s, c, b, t) = (0.00216, 0.00470, 0.0935, 1.273, 4.183, 172.3523553288311) \text{ GeV}$$

on `current_family_only`, and the restricted common-refinement transport-frame theorem reconstructs the same sextet and explicit exact forward Yukawas on `current_family_common_refinement_transport_frame_only`. Those supporting surfaces agree with the selected-class support wrapper on the exact running target surface; they do not prove the missing source sigma selector.

A separate D12 internal backread sidecar is on disk. On that continuation-only surface, feeding the emitted reference-free forward light-quark pair through the coefficient-ratio identification $c_u/c_d = y_u/y_d = m_u/m_d$ and the D12 normalization $\Delta_{ud}^{\text{overlap}} = \frac{1}{6} \log(c_d/c_u)$ fixes

$$\begin{aligned}
\Delta_{ud}^{\text{overlap}} &= 0.13431740757663183, \\
t_1 &= 0.6715870378831591, \\
\eta_Q^{\text{centered}} &= -0.018210067243314802, \\
\kappa_Q &= -0.012436796997836279.
\end{aligned}$$

This closes the D12 mass-side scalar package numerically on that sidecar surface. It does not replace the selected-class public theorem and it does not classify all public quark frame classes.

8.4 Quark Theorem Boundary

The conditional downstream closure route is

$$f_P + \Sigma_{ud}^{\text{source}}(P) \implies (\sigma_u, \sigma_d, \sigma_{\text{seed}}^{ud}, \eta_{ud})_{\text{phys}} \implies (g_u, g_d) \implies (m_u, m_d, m_s, m_c, m_b, m_t),$$

with the exact forward construction emitting Y_u and Y_d on the same selected-class surface. The current artifact supplies a target-derived sigma witness on this route, so it is selected-class conditional support only. It does not classify all public quark frame classes, and it does not promote the quark masses until $P \rightarrow \Sigma_{ud}^{\text{source}}(P)$ is proved.

8.5 Strong-CP branch

The selected-class quark support wrapper conditionally carries the running sextet and explicit forward Yukawas on the public quark frame class f_P , subject to the missing source sigma selector. Strong CP is a separate phase-side invariant. The available corpus does not derive the bare QCD angle θ_{QCD} , does not emit the physical anomaly-invariant combination $\bar{\theta}$, and does not prove that the physical strong-CP phase vanishes.

The unfinished object is narrower than the quark mass theorem. The required bridge is a theorem-grade descent from exact quark/Yukawa phase data to the determinant-line phase contribution, together with a theorem fixing the topological-angle contribution on the realized branch.

9 Charged-Lepton Family Derivation

The charged-lepton derivation differs from the quark derivation in a precise way. It does not emit public charged masses from P . It emits an exact same-family witness, a closed common-shift no-go, the declared same-label q_e readback, a determinant-line lift on theorem-grade physical charged data, and a downstream algebraic mass readout from theorem-grade $A_{\text{ch}}(P)$. For any fixed formal source exponent vector $M_{\bullet}^{\text{ch}}$, the same-label readback defines a source-side determinant character

$$S_M = \sum_e M_e^{\text{ch}} \log q_e.$$

The theorem lane does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The electron, muon, and tau rows are therefore reported as n/a on the public theorem lane.

The mathematical split is equally precise. The charged-lepton derivation starts from the ordered charged package, proves that the realized support is a one-dimensional linear subray, exposes the canonical quadratic support-extension direction, maps that into the charged excitation gaps, closes a two-scalar support-extension law shell, isolates the smaller eta source-readback primitive on that same carrier, and then builds the log-spectrum and forward shape/scale surface. Those centered objects are common-shift invariant. The determinant line fixes the physical affine scalar once theorem-grade physical charged data are present, and the mass readout is then algebraic. The charged theorem boundary has two distinct pieces. One piece is promotion of the latent charged sector-response candidate to theorem-grade $\hat{C}_e$. The other is source-to-physical determinant attachment. For a fixed formal source exponent vector $M_{\bullet}^{\text{ch}}$, that attachment is the identity

$$3\mu(r) = \sum_e M_e^{\text{ch}} \log q_e(r),$$

equivalently zero determinant-normalization defect

$$N_{\det}(P) = s_{\det}(P) - \sum_e M_e^{\text{ch}} \log q_e(P),$$

on the charged determinant channel.

Physically, the electron is the light charged lepton that makes atoms and chemistry possible, the muon is its heavier unstable cousin, and the tau lepton is the heaviest charged lepton. The role of the charged-lepton chapter is to explain how those three rows emerge from the shared flavor dictionary without Koide-assisted fitting. The theorem-grade lane has to derive promotion of the latent charged sector-response candidate $\widehat{C}_e^{\text{cand}}$ to theorem-grade $\widehat{C}_e$ by closing the branch-generator splitting theorem. On theorem-grade physical Y_e , a refinement-stable uncentered lift collapses the determinant-line section and affine anchor to one descended physical affine scalar $\mu_{\text{phys}}(Y_e)$, with

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\text{phys}}(Y_e) \mathbf{1}, \quad s_{\det}(Y_e) = 3\mu_{\text{phys}}(Y_e), \quad A_{\text{ch}}(Y_e) = \mu_{\text{phys}}(Y_e).$$

Theorem 9.1 (Charged same-carrier source-pair readback). *Fix the ordered charged carrier*

$$\begin{aligned} &(-1, x_2, 1), \\ &x_2 = -0.5175863354681689. \end{aligned}$$

If the charged source pair

$$(\eta_{\text{ext}}, \sigma_{\text{ext}}) = (\eta_{\text{source_support_extension_log_per_side}}, \sigma_{\text{source_support_extension_total_log_per_side}})$$

is emitted on that carrier, then the centered charged logs are

$$\begin{aligned} e_{\log, \text{centered}} &= -\frac{(3 + x_2)\sigma_{\text{ext}} - \eta_{\text{ext}}}{6}, \\ \mu_{\log, \text{centered}} &= \frac{x_2\sigma_{\text{ext}} - \eta_{\text{ext}}}{3}, \\ \tau_{\log, \text{centered}} &= \frac{(3 - x_2)\sigma_{\text{ext}} + \eta_{\text{ext}}}{6}, \end{aligned}$$

and therefore the charged masses are

$$m_e = g_e e^{e_{\log, \text{centered}}}, \quad m_\mu = g_e e^{\mu_{\log, \text{centered}}}, \quad m_\tau = g_e e^{\tau_{\log, \text{centered}}}.$$

This theorem is exact on the same-carrier shell, but the absolute values are not emitted. The visible scalar order is

$$\eta_{\text{ext}} \quad \text{then} \quad \sigma_{\text{ext}},$$

and the charged absolute scale g_e is unresolved. Builder-facing code therefore exposes η_{ext} and then σ_{ext} as the first same-carrier residuals, but the theorem-grade boundary lies above that surface. If the latent candidate $\widehat{C}_e^{\text{cand}}$ is promoted, then η_{ext} and σ_{ext} become charged spectral invariants instead of separate primitive goals, and the absolute-scale burden is pushed to one affine-covariant absolute charged anchor A_{ch} . In the local chain, $\widehat{C}_e$ itself is undeclared: only the centered compressed generation-bundle branch-operator candidate $\widehat{C}_e^{\text{cand}}$ is on disk, and the operator-side gate for that package is the upstream promotion theorem `oph_generation_bundle_branch_generator_splitting`, not a new ad hoc charged operator ansatz. Its exact gate clause is the compression-descendant commutator statement `compression_descendant_commutator_vanishes_or_is_uniformly_quadratic_small_after_central_split`. The centered common-shift quotient is closed negatively, so centered data alone do not emit the affine anchor A_{ch} .

Theorem 9.2 (Charged absolute-scale underdetermination). *Let*

$$E_e^{\text{centered}} = (e_{\log, \text{centered}}, \mu_{\log, \text{centered}}, \tau_{\log, \text{centered}})$$

be the centered charged log triple emitted from the charged source pair $(\eta_{\text{ext}}, \sigma_{\text{ext}})$. Then for every $c \in \mathbb{R}$,

$$Y_e(c) := \exp(c) \text{diag}(e^{E_e^{\text{centered}}})$$

has the same charged spectral invariants

$$\eta_{\text{ext}}, \quad \sigma_{\text{ext}}, \quad \gamma_{21}, \quad \gamma_{32},$$

the same centered log vector, and the same ratio data. Only the absolute masses scale:

$$(m_e, m_\mu, m_\tau) \mapsto e^c (m_e, m_\mu, m_\tau).$$

Hence the charged OPH chain determines only the quotient class

$$E_e^{\text{centered}} \in \mathbb{R}^3 / \langle (1, 1, 1) \rangle,$$

not the absolute scalar $g_e = e^c$.

The proof is immediate from the centered sum rule

$$e_{\log, \text{centered}} + \mu_{\log, \text{centered}} + \tau_{\log, \text{centered}} = 0,$$

which implies

$$\det Y_e^{\text{shape}} = 1, \quad \det Y_e = g_e^3.$$

All emitted charged invariants depend only on differences of log entries, so a common shift in the $(1, 1, 1)$ direction leaves them unchanged. This is the exact reason the available corpus closes the P -driven charged lane as a no-go instead of as a mass theorem: the theorem surface fixes the centered shape, while the determinant-line landing fixes the absolute normalization.

The charged absolute-scale lane is explicitly typed. The charged scale is a linear quantity

$$g_e = e^{\mu_e^{\text{abs}}},$$

so the log-coordinate μ_e^{abs} must not be mixed directly with centered log gaps. The same-family writeback therefore records the type-consistent shell

$$\begin{aligned} \mu_{e, \text{seed}}^{\text{abs}} &= \log(0.9231656602589082) \\ &= -0.07994658034676537, \end{aligned}$$

$$\mu_{e, \text{cand}}^{\text{abs}} = \mu_{e, \text{seed}}^{\text{abs}} - \gamma_{\min} = -0.38231224060567365,$$

$$g_e^{\text{cand}} = e^{\mu_{e, \text{cand}}^{\text{abs}}} = 0.6822819838027987.$$

This is a representation-consistency shell only, not a charged-mass theorem. It fixes the linear-vs-log coordinate discipline. This surface does not emit a theorem-grade value law for g_e . The audited shortcut

$$\begin{aligned} \Delta_e^{\text{abs}} &= 0.30236566025890826, \\ g_e &= 0.6822819838027987 \end{aligned}$$

is not a theorem-grade closure. It merely chooses one representative on the common-shift orbit and does not land on the physical charged masses.

The next theorem object beyond this shell is one theorem-grade affine-covariant section A_{ch} of the quotient map, satisfying

$$A_{\text{ch}}(\ell + c\mathbf{1}) = A_{\text{ch}}(\ell) + c.$$

The clean realization of that section is an uncentered charged response lift carrying a determinant line, in which

$$A_{\text{ch}} = \frac{1}{3} \log \det(Y_e) = \frac{1}{3} \text{tr}(\log Y_e).$$

Once such an anchor exists,

$$g_e = e^{A_{\text{ch}}(\ell)}, \quad \Delta_e^{\text{abs}} = \log g_{\text{ch}}^{\text{shared}} - A_{\text{ch}}(\ell).$$

Compare-only charged continuation bridge. For comparison, there is a sharper charged continuation bridge under three extra continuation assumptions:

1. a uniform $\mathbb{Z}_6$ center-label ensemble, so $\varepsilon = 1/6$;
2. the balanced Hermitian circulant branch, so $|b|/a = 1/\sqrt{2}$;
3. the OPH phase choice $\delta = 2/9$.

With the scale-free normalization $a = 1$, the ordered roots

$$r_k = a + 2|b| \cos\left(\delta + \frac{2\pi k}{3}\right)$$

become

$$\begin{aligned} r_e &= 0.040349908219207475, \\ r_\mu &= 0.5802119201475368, \\ r_\tau &= 2.3794381716332555. \end{aligned}$$

Under those assumptions the bridge selects the same-carrier pair

$$\begin{aligned} \eta_{\text{ext}} &= -6.729586682888832, \\ \sigma_{\text{ext}} &= 8.154061112725994, \end{aligned}$$

with ordered gap values

$$\begin{aligned} \gamma_{21} &= 5.33160859254774, \\ \gamma_{32} &= 2.822452520178255, \\ \kappa_{\text{ext}} &= -4.59605680397234, \end{aligned}$$

and centered logs

$$E_{\log, \text{centered}} = [-4.495223235091244, 0.836385357456495, 3.6588378776347503].$$

Against the charged references, this continuation-centered shape has residual norm

$$\left\| E_{\log, \text{centered}}^{\text{cont}} - E_{\log, \text{centered}}^{\text{ref}} \right\| \approx 2.13 \times 10^{-5},$$

so this compare-only branch is a near-exact centered-shape closure up to one common absolute scale. The theorem lane remains unpromoted because the public affine absolute anchor is external

to this near-exact centered charged-shape branch. The compare-only common scale required for exact absolute masses is

$$g_e^* = 0.04577885783568762.$$

Equivalently, relative to the stored shared-budget seed

$$g_{\text{ch}}^{\text{shared}} = 0.9231656602589082,$$

the missing affine absolute anchor on the determinant-line route would have to take the target value

$$\Delta_e^{\text{abs},*} = \log \frac{g_{\text{ch}}^{\text{shared}}}{g_e^*} = 3.003986333402356.$$

This identifies the public charged theorem boundary: the compare-only branch nearly solves the centered charged shape, but the theorem-grade derivation lacks promotion of the latent candidate $\hat{C}_e^{\text{cand}}$ to theorem-grade $\hat{C}_e$, and then one affine-covariant absolute anchor A_{ch} that would turn that centered readback into public charged masses on the theorem lane.

10 Neutrino Family Derivation

The neutrino lane contains one exact no-go and one failed comparison candidate. The intrinsic isotropic branch is excluded by its spectral cap. The weighted-cycle construction descends from two hand-written family-transport matrices and declared label, orientation, cycle, and exponent choices. Its exponent law was promoted after candidate laws had been ranked against the measured splitting ratio. The resulting point therefore has retrospective, target-informed template status. The bridge and absolute attachment use compare-only information and carry no prediction status.

For the isotropic matrix $M = aI + \rho C$ with unit-modulus off-diagonal entries, the general Gershgorin estimate applied to $M^\dagger M$ gives

$$\max_{i,j} |\Delta m_{ij}^2| \leq 8a\rho + 4\rho^2.$$

The live bound is $1.52304 \times 10^{-6} \text{ eV}^2$, still about three orders of magnitude below the atmospheric scale. The previously quoted $8a\rho$ expression omitted the $4\rho^2$ term and was not a valid general bound.

The frozen weighted-cycle candidate gives

$$\theta_{12} = 34.225904631810025^\circ,$$

$$\theta_{23} = 49.72282845058266^\circ,$$

$$\theta_{13} = 8.686355527700156^\circ,$$

$$\delta_{\text{PMNS}} = 305.58061231449796^\circ,$$

$$J = -0.02753115613565372,$$

and the dimensionless hierarchy invariant

$$\frac{\Delta m_{21}^2}{\Delta m_{32}^2} = 0.030721110097966534.$$

The official NuFIT 6.1 normal-ordering profile gives

$$\Delta\chi_{23,\delta}^2 = 20.11955 \quad \text{with the tabulated atmospheric likelihood,} \quad \Delta\chi_{23,\delta}^2 = 18.43528 \quad \text{without it,}$$

at $(\sin^2 \theta_{23}, \delta_{\text{CP}}) = (0.582056, -54.419^\circ)$ [9]. Both values exceed the two-parameter 3σ contour value 11.83. Separate marginal interval checks had hidden this correlation. The published profiles overlap and are not summed; their maximum is a lower bound on the unavailable full fixed-candidate $\Delta\chi^2$.

The historical shared-basis construction set

$$M_{\text{shared}} = U_{e,\text{left}}^* M_{\text{wc}} U_{e,\text{left}}^\dagger, \quad U_{\nu,\text{shared}} = U_{e,\text{left}} U_{\text{wc}},$$

the identity

$$U_{e,\text{left}}^\dagger U_{\nu,\text{shared}} = U_{\text{wc}}.$$

This cancellation is a tautology because $U_{\nu,\text{shared}}$ was defined as $U_{e,\text{left}} U_{\text{wc}}$. It does not independently identify the physical charged-lepton basis or validate the PMNS point. No theorem in the current stack places the f -labelled weighted-cycle operator in the charged-lepton mass basis.

The stored charged-basis artifact also fails its own promotion gate. Its source has `closure_state: open`, and its three normalized shape singular values differ only at about the 10^{-12} relative level, so its left singular vectors do not define a stable physical family basis. The newer charged-lepton continuation remains support-extension gated and does not supply a replacement labelled basis. At source level, the physical PMNS matrix is therefore unformed.

Taking the stored basis declarations literally gives a sharply different diagnostic. Reordering M_{wc} from $[f3, f1, f2]$ to the independently declared shared basis $[f1, f2, f3]$, computing its Takagi matrix $U_\nu^{(f)}$, and then forming $U_{e,\text{left}}^\dagger U_\nu^{(f)}$ yields

$$\theta_{12} = 38.6812^\circ, \quad \theta_{23} = 76.2759^\circ, \quad \theta_{13} = 44.8283^\circ,$$

with $\delta = 324.2140^\circ$ and $J = -0.0233164$. In particular, $\sin^2 \theta_{13} = 0.4970$ and $\sin^2 \theta_{23} = 0.9437$, both outside the corresponding NuFIT 6.1 profile grids. This is a diagnostic conditional on the current stored matrices and label order. It is not a source-closed prediction because the charged basis and family kernel also descend from template-level inputs.

The correct transported Takagi identity is

$$U_{\nu,\text{shared}}^T M_{\text{shared}} U_{\nu,\text{shared}} = \text{diag}(m_i) \in \mathbb{R}_{>0}.$$

Here U_{wc} denotes the canonical phase-fixed Takagi unitary, not the raw arbitrary-phase eigensystem of $M_{\text{wc}}^\dagger M_{\text{wc}}$. Equivalently, $U_{\text{wc}}^T M_{\text{wc}} U_{\text{wc}}$ is positive diagonal in the basis where M_{wc} was declared. The earlier expression with $U_{\text{PMNS}}^T M_{\text{shared}} U_{\text{PMNS}}$ was algebraically false. Canonical Takagi readout under the declared charged-basis assumption and the electron-row gauge $U_{e1} \in \mathbb{R}_{>0}$ gives the comparison-only Majorana pair

$$\alpha_{21}^{(\text{Maj})} = 153.6185177794357^\circ, \quad \alpha_{31}^{(\text{Maj})} = 257.0032408220805^\circ.$$

The declared exponent law uses the positive transport-load segment between $\chi = 1 + \epsilon$ and $1 + \gamma_{1/2}$: on a one-dimensional affine segment, the balanced selector and the least-distortion selector for any positive translation-invariant quadratic form coincide at the midpoint, so

$$D_\nu = \frac{\chi + (1 + \gamma_{1/2})}{2}, \quad p = 1 + \gamma + \frac{\epsilon}{D_\nu}.$$

The midpoint fact does not derive the segment endpoints, the exponent law, the cycle topology, or their application to neutrino transport. Git history records target-ranked model selection before this law was promoted. The formula is therefore part of the frozen candidate rather than a source-side

consequence of OPH. On the declared positive selector segment there is a stronger compare-only two-parameter adapter: under the declared normal-ordering hypothesis, solving τ_ν against the representative central ratio and then solving λ_ν against Δm_{32}^2 gives

$$(m_1, m_2, m_3) = (0.01745663295, 0.01948419960, 0.05308139066) \text{ eV},$$

$$\begin{aligned}\Delta m_{21}^2 &= 7.49 \times 10^{-5} \text{ eV}^2, \\ \Delta m_{31}^2 &= 2.5129 \times 10^{-3} \text{ eV}^2, \\ \Delta m_{32}^2 &= 2.438 \times 10^{-3} \text{ eV}^2.\end{aligned}$$

These exact central numbers are compare-only. The bridge audit also found target feedback in the reduced invariant, so the following values are retained only as diagnostic coordinates:

$$\begin{aligned}C_\nu &= 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337,\end{aligned}$$

and therefore

$$\begin{aligned}\lambda_\nu &= \frac{m_{\star, \text{eV}}}{q_{\text{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \quad \Delta m_{ij}^2 = \lambda_\nu^2 \hat{\Delta}_{ij}.\end{aligned}$$

The normalized same-label overlap-defect calculation is exact after the template and declared selectors are supplied. The positive-segment adapter, bridge corridor, and bridge-coordinate sidecars remain diagnostic surfaces. In particular, the following quantity is only a diagnostic coordinate:

$$C_\nu := \frac{B_\nu}{I_\nu^{1/2} \widehat{\text{ratio}}^{1/2} \text{sum_defect}^{-1}}$$

It was defined above the declared proxy and normalizer. The attached stack also factors exactly through $q_e = q_{\text{mean}} q_{\text{bar}_e}$, so a q_{bar_e} -only collapse law for the bridge factor does not follow from the conditional algebraic surface stated here. The best algebraically complete conditional local object beneath that bridge is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with the induced μ_e family, but that family sits below C_ν and the induced paper-facing amplitude B_ν instead of replacing them. No source-only PMNS, hierarchy, Majorana-phase, or absolute-mass prediction survives these gates.

Cosmological-neutrino pressure surface. The compare-only absolute attachment displays

$$\sum_i m_{\nu_i} = 0.09001192964464505 \text{ eV}.$$

Under the declared normal-ordering hypothesis, standard relic-neutrino inheritance, and absent an explicitly declared extra OPH relativistic coherence channel, the diagnostic cosmological-neutrino-background branch uses

$$N_{\text{eff}}^{\text{OPH}} = 3.044,$$

with the usual broad, low-amplitude normal-ordering free-streaming imprint. This is a diagnostic cosmological pressure surface for the rejected candidate. The displayed mass sum sits below the Planck+BAO 0.12 eV bound [7] but is exposed to strict DESI DR2 Λ CDM-style bounds near 0.0642 eV under their model assumptions [8]. A cosmological upper bound below the displayed

compare-only value would exclude the displayed absolute-mass continuation under a model class that also respects oscillation lower limits and the OPH background assumptions.

On the same-label neutrino-only branch, the centered eta-class is exactly S_3 -isotropic, so the same-label data are edge-constant and the solar 1–2 split cannot open there by itself. The first solar mover therefore has to come from realized flavor-side same-label gap/defect readback, not from another neutrino-only selector tweak.

10.1 Intrinsic neutrino eta-chain

The proof-facing input is smaller than the raw same-label matrix payload. It compresses to the same-label scalar certificate

$$(g_e, \omega_e)_{e \in \{12, 23, 31\}}, \quad \omega_e = \text{same-label overlap}_e^2, \quad d_e = 1 - \omega_e,$$

modulo one common scale. From that certificate one forms

$$q_e = \sqrt{g_e d_e}, \quad \eta_e = \log q_e - \frac{1}{3} \sum_f \log q_f, \quad e \in \{12, 23, 31\},$$

the intrinsic selector depends only on the centered class

$$[\eta_e] \in \mathbb{R}^3 / \langle (1, 1, 1) \rangle.$$

Equivalently one may work with any positive normalized family

$$\mu_e = \frac{e^{\eta_e}}{\frac{1}{3} \sum_f e^{\eta_f}}.$$

Common scaling cancels identically, so the intrinsic selector is determined by the centered eta-class alone.

This factorization result is conditional on the scalar inputs; it is not a source-closure result. The current scalar values are numerically complete, but their gap data inherit the template family-transport kernel and their overlap data inherit a candidate-only line lift. The code now propagates those upstream statuses, so the intrinsic spectrum remains a conditional diagnostic until both inputs close at source level.

On the isotropic neutrino-only branch one has

$$(\eta_{12}, \eta_{23}, \eta_{31}) = (0, 0, 0),$$

which gives the conditional intrinsic singular values

$$(s_0, s_1, s_2) = (2.3986448447627196, 2.3986448447627196, 2.590074050773907) \times 10^{-12} \text{ GeV},$$

with ascending squared-mass gaps

$$\begin{aligned} s_1^2 - s_0^2 &= 0, \\ s_2^2 - s_0^2 &= 9.549864971855843 \times 10^{-25} \text{ GeV}^2. \end{aligned}$$

This is the exact neutrino-only isotropy obstruction behind the solar-splitting boundary.

Theorem 10.1 (Fixed isotropic reference plus same-label scalars suffice for the intrinsic deformation). *Fix the isotropic reference M_0 , equivalently its parameters (a, ρ, Ω) . Assume the same-label gap and overlap scalars are supplied on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e, \omega_e)_{e \in \{12, 23, 31\}}$$

or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ascending gaps, or the intrinsic spectral subspaces. When the singular spectrum is simple, canonical Takagi congruence fixes an ordered column basis up to column signs. At a degeneracy, including the isotropic $s_0 = s_1$ point, only the degenerate spectral subspace and its projector are fixed. Ascending singular-value sorting does not select the physical normal or inverted mass-eigenstate labels.

Theorem 10.2 (Exact principal-branch selector from the centered eta-class). *Fix the cycle sum*

$$\Omega = \psi_{12} + \psi_{23} + \psi_{31},$$

and a positive weight family μ_e . Define

$$\mu_{\min} = \min_e \mu_e, \quad F_{\max} = \sum_e \arcsin\left(\frac{\mu_{\min}}{\mu_e}\right).$$

If $|\Omega| < F_{\max}$, then on the principal branch $\psi_e \in (-\pi/2, \pi/2)$, the affine energy

$$A(\psi) = \sum_e \mu_e (1 - \cos \psi_e)$$

has a unique minimizer. It is given by

$$\psi_e^* = \arcsin(\lambda/\mu_e),$$

where λ is the unique solution of

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \quad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

Proof. The Euler–Lagrange equations are $\mu_e \sin \psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range on $(-\mu_{\min}, \mu_{\min})$ is $(-F_{\max}, F_{\max})$, so $F(\lambda) = \Omega$ has a unique solution under the stated condition. Strict convexity follows because the Hessian of A is

$$\nabla^2 A = \text{diag}(\mu_e \cos \psi_e),$$

which is positive definite on the principal branch and is positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. $\square$

For the live normalized family

$$(\mu_{12}, \mu_{23}, \mu_{31}) = (1.4497003, 0.9968526, 0.5534471),$$

one has $F_{\max} = 2.5511001$, $|\Omega| = 1.7561443$, and positive domain margin 0.7949558. The corrected theorem domain therefore leaves the stored numerical selector unchanged.

Let

$$a = m_{\star} = \frac{v^2}{\mu_u}, \quad \rho = |(M_0)_{12}|.$$

Then the intrinsic Majorana matrix is

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

Theorem 10.3 (Exact intrinsic spectral cubic). *Let $H = M^\dagger M$. Then*

$$H = dI + T, \quad d = a^2 + 2\rho^2,$$

where T has zero diagonal and off-diagonals

$$x_{12} = 2a\rho \cos \psi_{12} + \rho^2 e^{i(\psi_{23} - \psi_{31})},$$

$$x_{23} = 2a\rho \cos \psi_{23} + \rho^2 e^{i(\psi_{31} - \psi_{12})},$$

$$x_{13} = 2a\rho \cos \psi_{31} + \rho^2 e^{i(\psi_{23} - \psi_{12})}.$$

Define

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \quad Q = \Re(x_{12}x_{23}\overline{x_{13}}).$$

Then the eigenvalues λ_k of T are exactly the three real roots of

$$\lambda^3 - P\lambda - 2Q = 0,$$

and the intrinsic squared masses are

$$m_k^2 = d + \lambda_k.$$

Equivalently,

$$\lambda_k = 2\sqrt{\frac{P}{3}} \cos\left(\frac{1}{3} \arccos\left(\frac{3\sqrt{3}Q}{P^{3/2}}\right) - \frac{2\pi k}{3}\right).$$

Corollary 10.4 (Intrinsic eta-chain spectral closure). *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$s_0 \leq s_1 \leq s_2, \quad s_1^2 - s_0^2, \quad s_2^2 - s_0^2, \quad s_2^2 - s_1^2,$$

and the corresponding intrinsic spectral subspaces with no PMNS import and no flavor-label leakage. A columnwise Takagi basis is emitted only on the simple-spectrum locus. The physical normal-ordering assignment is $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$, while the inverted-ordering assignment is $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The current source does not choose between them, so physical ordering remains open.

The matrix used in this corollary is Majorana, so its physical column matrix is the Takagi matrix U_ν defined by

$$U_\nu^T M U_\nu = \text{diag}(s_0, s_1, s_2) \in \mathbb{R}_{\geq 0}.$$

Earlier artifacts exported the left SVD matrix, which diagonalizes $MM^\dagger$, instead of the Takagi matrix, whose columns diagonalize $M^\dagger M$. The corrected congruence has maximum off-diagonal residual 5.4×10^{-28} GeV on the live anisotropic matrix. Conditional on the stale stored charged matrix, the corrected intrinsic combination would give

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (45.0137^\circ, 2.50489^\circ, 2.53597^\circ, 210.6251^\circ).$$

That diagnostic is strongly incompatible with the NuFIT angle surface, but it is not an OPH prediction because the charged basis is open and template-derived.

10.2 Perturbative laws around the isotropic point

Write

$$\psi_e = \varphi + \delta_e, \quad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan \varphi \eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3} \sum_e \delta_e^2.$$

Assume $2a \cos \varphi + \rho \neq 0$ and that the collective state is separated from the isotropic doublet. Then the two ascending doublet eigenvalues satisfy

$$s_{0,1}^2 = m_d^2 \mp 2a\rho |\sin \varphi| \sigma + O(\delta^2),$$

so

$$s_1^2 - s_0^2 = 4a\rho |\sin \varphi| \sigma + O(\delta^2) = 4a\rho \frac{\sin^2 \varphi}{|\cos \varphi|} \sqrt{\frac{2}{3} \sum_e \eta_e^2} + O(\eta^2).$$

Under the declared normal-ordering hypothesis, the ascending atmospheric gaps obey

$$\Delta m_{31}^2 = \Delta_{\text{atm,iso}} + \frac{1}{2} \Delta m_{21}^2 + O(\delta^2), \quad \Delta m_{32}^2 = \Delta_{\text{atm,iso}} - \frac{1}{2} \Delta m_{21}^2 + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\text{cent}} = s_2^2 - \frac{1}{2}(s_0^2 + s_1^2) = \Delta_{\text{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta \Delta_{\text{cent}} = -a\rho \sigma^2 \frac{a(4 \cos^2 \varphi - 1) + 6\rho \cos \varphi}{2a \cos \varphi + \rho} + O(\delta^3).$$

The projective largest-mass right-singular direction, in the phase gauge that approaches the real democratic vector at the isotropic point, obeys the first-order deformation law

$$u_3 = u + \kappa \begin{pmatrix} \delta_{23} \\ \delta_{31} \\ \delta_{12} \end{pmatrix} + O(\delta^2), \quad u = \frac{1}{\sqrt{3}}(1, 1, 1)^T, \quad \kappa = \frac{\sqrt{3}(2a \sin \varphi + 3i\rho)}{9(2a \cos \varphi + \rho)}.$$

This equation fixes only the complex line. If v_3 denotes its normalized right-hand side, the canonical Takagi representative used by the executable pipeline is

$$u_3^{\text{Tak}} = e^{-\frac{i}{2} \arg(v_3^T M_\nu v_3)} v_3, \quad (u_3^{\text{Tak}})^T M_\nu u_3^{\text{Tak}} = s_3 > 0,$$

up to the unavoidable column sign. In particular, at the isotropic point the real democratic vector is generally not already in positive Takagi gauge. Equivalently,

$$u_3 = u - \tan \varphi \kappa \begin{pmatrix} \eta_{23} \\ \eta_{31} \\ \eta_{12} \end{pmatrix} + O(\eta^2).$$

One exact demonstration uses centered eta-class

$$(\eta_{12}, \eta_{23}, \eta_{31}) = (0.13631072512014578, -0.18362987264261327, 0.0473191475224675),$$

and returns the ascending intrinsic singular values

$$(s_0, s_1, s_2) = (2.3929601069646055, 2.4048200109774875, 2.589606227283229) \times 10^{-12} \text{ GeV},$$

with

$$\begin{aligned} s_1^2 - s_0^2 &= 5.690121167370743 \times 10^{-26} \text{ GeV}^2, \\ s_2^2 - s_0^2 &= 9.798023388600230 \times 10^{-25} \text{ GeV}^2. \end{aligned}$$

These are exact outputs of the intrinsic eta-chain once the fixed isotropic reference and centered eta-class are supplied. They are not flavor-labeled OPH rows, and ascending sorting does not pick a physical ordering. Under the declared normal-ordering hypothesis $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$; under the inverted-ordering hypothesis $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The source-side label rule is absent.

10.3 Weighted-cycle bridge rigidity and absolute attachment

The weighted-cycle route supplies a frozen PMNS/hierarchy candidate. The reduced invariant C_ν was selected on a compare-only correction surface, with $B_\nu = P_\nu C_\nu$ retained as a diagnostic amplitude parameterization:

$$\begin{aligned} C_\nu &= G^2 Q S^{-1/2} = 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337. \end{aligned}$$

The blocked absolute attachment displays

$$\begin{aligned} \lambda_\nu &= \frac{m_{\star, \text{eV}}}{q_{\text{mean}}^{\nu}} P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \\ \Delta m_{ij}^2 &= \lambda_\nu^2 \widehat{\Delta m_{ij}^2}. \end{aligned}$$

The basis permutation, holonomy orientation, and CP sign lack source-side selection. Exhaustive enumeration of both stored orientations and all row and mass-column relabelings of the stored candidate matrix finds no normal-ordering-consistent relabeling that passes both NuFIT 6.1 correlated 3σ gates. This excludes a convention-only rescue of that matrix; it neither derives nor exhausts possible source-side physical charged-basis placements. The one-parameter atmospheric-anchor slice and the two-parameter positive-segment adapter remain compare-only continuations.

11 Hadrons, QCD, and the Emergence of Ordinary Matter

The hadron derivation is the most operationally demanding part of the particle program. The source-only artifact is not a fitted hadronic residual and not a bare scalar ρ_{had} . It is an OPH-QCD hadronic precision backend: a QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized electromagnetic current ledger, positive two-current spectral export $d\rho_Q^{(2)}$, higher-point and transition spectral exports, same-scheme remainder Ξ_Q , and systematics ledger. A source-only hadron mass row requires a working OPH hadron backend, such as GLORB/Echosahedron, that can emit these objects with manifest provenance and production systematics. Local surrogates, ordinary Chrome/Oracle workers, and bookkeeping scripts cannot promote this sector. The two-current spectral measure is the marginal needed for running- α and HVP transport; it is insufficient for HLbL and rare-decay long-distance amplitudes without the four-current and transition sectors. The source-backend boundary has an empirical closure policy. The empirical closure row class uses a separate $e^+e^- \rightarrow \text{hadrons}$ payload class for the hadronic spectral contribution and stays separate from source-only OPH rows. The source-only derivation nevertheless keeps the mathematical execution bridge, deterministic runtime receipt, writeback/evaluation path, and surrogate validation machinery as non-promoting scaffolding for the backend lane.

11.1 Seeded $2 + 1$ family and unquenched measure

Let

$$m_l := \frac{m_u + m_d}{2}, \quad \rho_l := \frac{m_u + m_d}{2 \Lambda_{\overline{\text{MS}}}^{(3)}}, \quad \rho_s := \frac{m_s}{\Lambda_{\overline{\text{MS}}}^{(3)}}.$$

The seeded fixed-physics family is

$$\begin{aligned} a\Lambda_n &= a\Lambda_{\text{seed}} 2^{-n}, & a\Lambda_{\text{seed}} &= \sqrt{\rho_l \rho_s}, \\ am_l^{(n)} &= a\Lambda_n \rho_l, & am_s^{(n)} &= a\Lambda_n \rho_s, \\ \beta_n &= 6 + \frac{9}{2\pi^2} \log \frac{1}{a\Lambda_n}, & L_n &= \left\lceil \frac{\lambda_L^{\text{target}}}{a\Lambda_n} \right\rceil, & T_n &= \left\lceil \frac{\lambda_T^{\text{target}}}{a\Lambda_n} \right\rceil. \end{aligned}$$

The unquenched ensemble measure is

$$d\mu_n(U) = Z_n^{-1} \exp[-S_g(U; \beta_n)] \det D_l(U; am_l^{(n)})^2 \det D_s(U; am_s^{(n)}) dU.$$

On this branch, $N_f = 2 + 1$ and QED is off.

11.2 Runtime receipt and deterministic cfg/source contract

The emitted execution law is

$$U_{n,c} = K_n^{N_{\text{therm}} + (\text{cfg_index}) N_{\text{sep}}} (U_{\text{cold}}; \text{seed}_{n,c}),$$

with stop-time formula

$$t_{\text{stop}}(n, c) = N_{\text{therm}} + (\text{cfg_index}) N_{\text{sep}}.$$

The deterministic cfg seed law is

$$\text{seed}_{n,c} = \text{bytes.fromhex}(\text{cfg_seed_hash}_{n,c}),$$

with

$$\text{cfg_seed_hash} = \text{SHA256}\left(\text{Serialize}(\text{ensemble_id}, \beta, L, T, am_l, am_s, \text{cfg_index})\right).$$

The source set is fixed as

$$S_{n,c} = \{[0, 0, 0, 0], [L_n//2, L_n//2, L_n//2, T_n//2]\}.$$

The non-null external runtime receipt used in the surrogate execution is

$$N_{\text{therm}} = 2048, \quad N_{\text{sep}} = 512.$$

These values are surrogate execution inputs only; they are not claimed as theorem outputs.

The production geometry summary makes the runtime boundary concrete. On the emitted seeded $2 + 1$ family there are three ensembles and six configurations total. If one stores all four links at every site as full double-complex 3×3 matrices, the naive raw gauge storage estimate is

$$576 \text{ bytes/site},$$

which gives total naive raw gauge storage

$$2.80071464105088 \times 10^{14} \text{ bytes}$$

over the production schedule. The normalized backend correlator dump needed by the analysis layer is tiny:

$$195264 \text{ bytes}$$

for the full π_{iso} , $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ payload on the production schedule. The hadron requirement is the backend export bundle that would feed the normalized dump from real production execution.

11.3 Operational reading of the required computation

It is useful to separate the lightweight analysis layer from the missing backend layer. Informally, the repository-side control plane is explicit: the seeded ensemble family, the deterministic cfg/source contract, the export manifest, the jackknife evaluator, the forward-window selector, and the publication-budget schema are all explicit. The missing step is the production export of Ward-projected hadronic spectral data from a real OPH hadron backend.

Technically, the required production computation is the standard lattice-QCD stable-channel workflow on the emitted family: run unquenched RHMC/HMC for the three seeded ensembles; for each realized cfg and fixed source solve the clover-improved Wilson light/strange systems; construct the zero-momentum π_{iso} , $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ two-point sequences; write the backend bundle; convert that bundle into the repo-side production dump; then let the existing jackknife and forward-window machinery emit $am_{X,\text{ground}}$, R_X , $m_X[\text{GeV}]$, and the published σ_{stat} , δ_{cont} , δ_{vol} , and δ_χ fields. This gives a complete source-only execution contract and a separate empirical display policy.

The engineering burden is therefore asymmetric. Local execution is sufficient for manifest generation, surrogate validation, and downstream readout tests, because those stages operate on a tiny correlator payload. The physical branch is different: it needs backend hardware and execution semantics that are absent from the repository environment. This paper therefore treats GLORB/Echosahedron-class OPH hardware, or an equivalent working OPH hadron backend, as the source-only backend requirement. Production hadron masses enter the particle pipeline only with a backend-emitted Ward-projected spectral measure and its uncertainty budget. This is a source-backend scope boundary.

11.4 Stable-channel correlators, effective masses, and forward windows

The pion stable channel is

$$p_\pi^{(n,c,s)}(t) = \sum_x \Re \text{tr}_{c,\text{spin}} [\gamma_5 S_l(x; s) \gamma_5 S_l(s; x)].$$

The nucleon stable channel is

$$p_{N,\text{dir}}^{(n,c,s)}(t) = \sum_x G_d, \quad p_{N,\text{ex}}^{(n,c,s)}(t) = \sum_x G_x,$$

$$p_N^{(n,c,s)}(t) = p_{N,\text{dir}}^{(n,c,s)}(t) - p_{N,\text{ex}}^{(n,c,s)}(t).$$

Cfg/source and ensemble averaging are

$$\bar{p}_X^{(n,c)}(t) = \frac{1}{|S_{n,c}|} \sum_{s \in S_{n,c}} p_X^{(n,c,s)}(t),$$

$$C_X^{(n)}(t) = \frac{1}{|C_n|} \sum_{c \in C_n} \bar{p}_X^{(n,c)}(t).$$

The effective-mass laws are

$$am_{\text{eff},\pi}(t) = \log \frac{C_\pi(t)}{C_\pi(t+1)}, \quad am_{\text{eff},N}(t) = \log \frac{|C_N(t)|}{|C_N(t+1)|}.$$

The forward window is

$$W_n = \{t : 1 \leq t+1 < \lfloor T_n/2 \rfloor\}.$$

The evaluator monitors log-convexity,

$$R_{\log \text{conv},\pi}(t) = C_\pi(t)^2 - C_\pi(t-1)C_\pi(t+1),$$

$$R_{\log \text{conv},N}(t) = |C_N(t)|^2 - |C_N(t-1)| |C_N(t+1)|,$$

tail-drop,

$$D_X(t) = am_{\text{eff},X}(t) - am_{\text{eff},X}(t+1),$$

and mirror suppression,

$$M_X(t) = \exp[-am_{\text{eff},X}(t)(T_n - 2t)].$$

The selected forward window is the longest contiguous run satisfying finite effective masses, non-negative log-convexity up to tolerance, nonnegative tail-drop up to tolerance, mirror suppression below threshold, and local plateau flatness. The candidate ground-state mass is then the weighted window average

$$am_{X,\text{ground}} = \frac{\sum_{t \in W_X^{\text{sel}}} w_t am_{\text{eff},X}(t)}{\sum_{t \in W_X^{\text{sel}}} w_t}, \quad w_t = \frac{1}{\max(\sigma_t^2, \varepsilon)}.$$

11.5 Statistics, systematics, and dimensional readout

Delete-1 jackknife is performed over the cfg axis after source averaging inside each cfg. With n_{cfg} configurations and integrated autocorrelation time $\tau_{\text{int,cfg}}$, the effective cfg count is

$$n_{\text{eff,cfg}} = \frac{n_{\text{cfg}}}{2\tau_{\text{int,cfg}}}.$$

The published statistical error is

$$\sigma_{\text{stat},X} = \text{JKstderr}(am_{X,\text{ground}}).$$

The machine-readable systematics field uses

$$\sigma_{\text{sys},X} = \sqrt{\delta_{\text{cont},X}^2 + \delta_{\text{vol},X}^2 + \delta_{\chi,X}^2}.$$

Continuum, volume, and chiral proxies are encoded as

$$\begin{aligned} R_X^{(n)} &= \frac{am_X^{(n)}}{a\Lambda_n}, & R_X^{(n)} &\approx R_X(0) + c_X(a\Lambda_n)^2, \\ \delta_{\text{cont},X}^{(n)} &= a\Lambda_n \left| R_X^{(n)} - R_X(0) \right|, \\ \delta_{\text{vol},\pi}^{(n)} &= am_\pi^{(n)} \frac{e^{-am_\pi^{(n)}L_n}}{\max(am_\pi^{(n)}L_n, 1)}, \\ \delta_{\text{vol},N}^{(n)} &= am_N^{(n)} \frac{e^{-am_\pi^{(n)}L_n}}{\max(am_\pi^{(n)}L_n, 1)}, \\ \delta_{\chi,\pi}^{(n)} &= a\Lambda_n \left| R_\pi^{(n)} - \langle R_\pi \rangle_n \right|, \\ Q_N^{(n)} &= \frac{R_N^{(n)}}{R_\pi^{(n)}}, & \delta_{\chi,N}^{(n)} &= a\Lambda_n \langle R_\pi \rangle_n \left| Q_N^{(n)} - \langle Q_N \rangle_n \right|. \end{aligned}$$

These are surrogate publication proxies, not production physical systematics. Given a ground-state candidate,

$$R_X = \frac{am_{X,\text{ground}}}{a\Lambda_{\overline{\text{MS}}}^{(3)}}, \quad m_X[\text{GeV}] = R_X \Lambda_{\overline{\text{MS}}}^{(3)}[\text{GeV}],$$

with

$$\Lambda_{\overline{\text{MS}}}^{(3)} = 0.3344017073 \text{ GeV}.$$

11.6 Surrogate execution bundle and frontier

The surrogate execution evolves a latent state $z \in \mathbb{R}^d$ with Hamiltonian

$$H(z, p) = S(z) + \frac{1}{2} \sum_i p_i^2,$$

where

$$S(z) = \frac{1}{2} \sum_i \omega_i^2 z_i^2 + \lambda_4 \sum_i z_i^4 + \kappa \sum_i (z_{i+1} - z_i)^2 + 2\alpha_l \sum_i \log(\mu_l^2 + z_i^2) + \alpha_s \sum_i \log(\mu_s^2 + \frac{1}{2} z_i^2).$$

Leapfrog integration plus Metropolis accept/reject gives the surrogate HMC update

$$(z, p) \mapsto (z', p'), \quad P_{\text{acc}} = \min(1, e^{-\Delta H}).$$

This kernel is a deterministic executable surrogate honoring the emitted receipt and seed law, with no physical lattice-QCD RHMC/HMC claim.

For validation of the execution bridge, the surrogate locks the ground-state masses to the hadron audit proxy values

$$m_{\pi, \text{proxy}} = 0.13497682776768472 \text{ GeV},$$

$$m_{N, \text{iso}, \text{proxy}} = \frac{m_p + m_n}{2} = 0.93891875434 \text{ GeV}.$$

Thus

$$am_{\pi, \text{proxy}}^{(n)} = \frac{m_{\pi, \text{proxy}}}{\Lambda_{\overline{\text{MS}}}^{(3)}[\text{GeV}]} a\Lambda_n,$$

$$am_{N, \text{iso}, \text{proxy}}^{(n)} = \frac{m_{N, \text{iso}, \text{proxy}}}{\Lambda_{\overline{\text{MS}}}^{(3)}[\text{GeV}]} a\Lambda_n.$$

The surrogate correlator is

$$C_X^{\text{sur}}(t) = A_{X,0} e^{-am_X t} + A_{X,1} e^{-am_{X, \text{ex}} t} + A_{X, \text{mir}} e^{-am_X (T_n - t)}$$

up to a multiplicative correlated noise factor. Direct and exchange nucleon pieces are written as

$$C_{N, \text{dir}}^{\text{sur}}(t) = f_{\text{dir}} C_N^{\text{sur}}(t), \quad C_{N, \text{ex}}^{\text{sur}}(t) = (f_{\text{dir}} - 1) C_N^{\text{sur}}(t),$$

so that

$$C_N^{\text{sur}}(t) = C_{N, \text{dir}}^{\text{sur}}(t) - C_{N, \text{ex}}^{\text{sur}}(t).$$

On the finest surrogate ensemble, the resulting diagnostic candidates are

$$m_{\pi, \text{iso}}^{\text{sur}} = 0.135039383836 \text{ GeV}, \quad m_{N, \text{iso}}^{\text{sur}} = 0.938960210578 \text{ GeV},$$

with worst absolute error approximately 6.26×10^{-5} GeV across the surrogate stable-channel family. These numbers validate the emitted execution bridge. They are not promotable production hadron predictions.

So the hadron derivation closes the full

receipt $\rightarrow$ execution $\rightarrow$ writeback $\rightarrow$ evaluation $\rightarrow$ budgets $\rightarrow$ forward-window summary

path on executed surrogate data, while the physical closure requires production unquenched RHMC/HMC, real Dirac solves and baryon contractions, production autocorrelation studies, and production continuum / finite-volume / chiral systematics.

12 Observer-Centric Particle Ontology and Measurement

This paper cannot stop at masses and couplings. OPH is observer-centric at the level of its basic ontology, so a particle chapter also has to explain what a particle *is* in this language and how a measurement turns an excitation surface into an actual observed state. Fortunately, this is one of the places where the OPH theorem surface contains real mathematical content instead of only interpretation.

12.1 Observer patches, overlap consistency, and particle data

Ref. [3] formulates the basic kinematic picture in patch-net language: each observer patch carries a local state, neighboring patches compare a shared interface alphabet, and a global state is physically admissible exactly when neighboring projections agree on the overlap. The algebraic OPH version of the same statement is Axioms 4.1 and 4.2: local physical data are carried by patch algebras $\mathcal{A}(P)$, and those local states must agree on shared subalgebras whenever patches overlap.

The particle interpretation starts from patch-local algebras, overlap-visible observables, edge sectors, and transport data rather than a single absolute global particle basis. A particle row in the reported derivation is a readout claim about a stable or candidate-stable excitation structure visible to a family of observer patches and consistent on their overlaps.

In the observer formulation, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is the observer patch, $\mathcal{A}(P)$ its local algebra, ρ the local state, and R the record algebra. For the particle derivation, R belongs to the quantum observer surface itself. On the exact fixed-cutoff measurement surface it is generated by central record projectors, and practical readout may use approximately commuting projectors that are close to that central reference algebra. That is what makes the observer-facing record surface shareable across overlapping observer descriptions without violating the usual no-cloning constraints on generic quantum states.

12.2 Record algebras and definite outcomes

The integrated measurement appendices and Ref. [4] make the measurement interface precise at fixed cutoff. On the declared operational surface, the completed write/verify slice carries a finite commutative central record algebra. Practical readout may instead use projectors Q_a on the same declared slots with commuting central reference projectors $\hat{Q}_a$, where

$$\delta_{\text{rec}} := \max_a \|Q_a - \hat{Q}_a\|.$$

Then

$$\|[Q_a, Q_b]\| \leq 4 \delta_{\text{rec}},$$

and, whenever $\|\tilde{\rho} - \rho\|_1 \leq \varepsilon$,

$$\left| \text{Tr}(\tilde{\rho} Q_a) - \text{Tr}(\rho \hat{Q}_a) \right| \leq \varepsilon + \delta_{\text{rec}}.$$

On the explicit fixed-cutoff screen architecture, the exact record algebra after a completed write/verify cycle is

$$\mathcal{Z}_{\text{rec}}(t) = \text{Alg}\left(\{P_{m_I^{(J)}(t)}\}_{I \neq J}, \{P_{r_I^{\text{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I < J, \alpha}\right).$$

Ref. [4] proves that, on the declared operational measurement surface, $\mathcal{Z}_{\text{rec}}(t)$ is commutative and central for the readout instrument being used.

This fixed-cutoff centrality result is what supports definite outcomes without adding a second ontology. The integrated measurement ledger phrases the same idea through edge-center decomposition: on a fixed collar with exact Markov structure, the state splits into superselection blocks,

$$\rho_{ABC} = \bigoplus_j q_j \rho_{Ab_L^{(j)}}^{(j)} \otimes \rho_{b_R^{(j)}C}^{(j)},$$

and the label j is classical center data. The supplement then goes one step further: in the physical algebra there are no interference observables between different sector blocks. So once a particle detection event has been recorded in the observer-accessible center/record algebra, the accessible physics is organized as a classical mixture over those blocks instead of as a superposed measurement record.

Accordingly, a measured particle family or excitation channel should be thought of as an observer-accessible sector or record value carried by the central measurement surface, not as a mysterious absolute collapse of the full universe-state into a preferred basis chosen by hand.

12.3 Born probabilities and post-measurement states

The fixed-cutoff measurement package is also explicit about probabilities and updates. For every event E in the sigma algebra generated by $\mathcal{Z}_{\text{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \text{Tr}(\rho_t P_E),$$

where $P_E \in \mathcal{Z}_{\text{rec}}(t)$ is the projector for that event. Conditioning on the event then produces the post-measurement state

$$\rho_t|_E = \frac{P_E \rho_t P_E}{\text{Tr}(\rho_t P_E)}.$$

In the supplement, the same statement appears as the Lüders update on the record algebra. In the synthesis paper *Observers Are All You Need* [2], this fixed-cutoff Born/Lüders package and its Bell/CHSH extension are imported as theorem-bearing, not non-theorem commentary.

This directly answers the particle-state question in the present framework. Measurements give particles actual states by conditioning the observer-accessible state on the central record algebra. A detector click, a stable overlap-sector readout, or a pointer value does more than announce a pre-existing classical label; it defines the conditioned state for subsequent observer-accessible physics. If the same event is re-read without any accepted repair move touching its support, the result in Ref. [4] shows that the re-read probability is 1. So the measurement interface is well defined and operationally auditable.

12.4 Particle identity as transport-stable structure

Once one asks “which event happened?” and then “which particle family is this?”, the flavor derivation becomes essential. The active flavor derivation does not start with the names electron, muon, tau lepton, or up quark built into the fundamental observable. It starts with transport kernels, generation-bundle data, same-label eigenline transport, overlap-edge cocycles, and a persistent flavor observable carrying intrinsic labels $f1, f2, f3$. This is an important discipline condition in the paper. At the deepest flavor surface used here, family identity is transport-stable intrinsic structure first and named experimental family assignment second.

The shared Yukawa/excitation dictionary carries substantial weight even though it is not itself the leading missing object. The flavor pipeline exports an invariant base: projectors, spectral gaps, pair suppressions, cycle phases, and common sector-response objects. The nonpromoting component is the map from that intrinsic base to the named low-energy fermion families when the required sector bridge is not emitted. The quark derivation emits named rows because the forward branch is far enough along to support direct numerical comparison there; the charged-lepton derivation does not do so at theorem grade. The neutrino weighted-cycle construction remains a target-informed template candidate and does not close on a physical flavor-labeled surface.

Accordingly, particle identity is a refinement-stable observer-accessible transport pattern whose named interpretation is inherited from the available dictionary and comparison surfaces. Where that dictionary is not emitted by the theorem surface, the paper states that boundary explicitly.

13 H3 Record Charts and Cross-Boundary Token Stitching

The previous subsection concerns sector and family identity: which transport-stable excitation type is being read. A different question is whether two localized record tokens, seen near a chart or partition boundary, are the same continuing observer-visible token. This section closes that second question at the finite-certificate level. It does not add a particle species, mass, charge, or scattering theorem. It says when a bounded patch system is allowed to stitch local H^3 proto-records into nonbranching record-worldlines, and when it must report ambiguity.

13.1 H3 chart and clock atlas

The existence, Lorentz naturality, and exact dimension of this atlas are imported from the compact cap-normal H^3 reconstruction theorem. That theorem identifies sky directions with $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, and maps each cap to a geodesic half-space in H^3 . The present section begins one gate later: it supplies record supports, observer clocks, chart transitions, gauge transport, and worldline stitching inside that chart. The cap-normal chart, cap-response localization, overlap descent, and worldline stitching are separate certificates.

Fix a curvature radius $R_H > 0$ and write

$$H_{R_H}^3 = \{X \in \mathbb{R}^{1,3} : \langle X, X \rangle_L = -R_H^2, X^0 > 0\}, \quad \langle X, Y \rangle_L = -X^0 Y^0 + X^1 Y^1 + X^2 Y^2 + X^3 Y^3.$$

The spatial distance used by the stitch certificate is

$$d_H(X, Y) = R_H \operatorname{arcosh} \left(-\frac{\langle X, Y \rangle_L}{R_H^2} \right).$$

An H^3 record atlas is a tuple

$$\mathcal{A}_H = (R_H, \{U_i, \chi_i, u_i, J_i, \vartheta_i\}, \{G_{ji}, F_{ji}\}, E, \nabla, \operatorname{Prov}_{R_H}, \operatorname{Prov}_u).$$

Here χ_i are $H_{R_H}^3$ charts, $u_i \in H_{R_H}^3$ is the selected local observer-frame anchor when such an anchor is claimed, J_i are chart domains for record support, and $\vartheta_i : J_i \rightarrow T$ are observer-clock maps into a common comparison-time line T . $\operatorname{Prov}_{R_H}$ records whether R_H is a unit convention, an imported physical scale, or an independently derived OPH scale; #309 supplies only the unit chart convention by itself. Prov_u records which observer, tetrad, or clock object selected the anchors u_i . The transition $G_{ji} \in \operatorname{SO}^+(1, 3)$ is the H^3 chart transition, while F_{ji} is the sector or gauge transport functor on the record payload. These two holonomies are deliberately separate: atlas holonomy tests geometry, and gauge holonomy tests transported sector data. When two charts describe the same transported observer frame, the atlas additionally requires

$$G_{ji} u_i = u_j.$$

Theorem 13.1 (H3 geometric naturality). *For every $A \in \operatorname{SO}^+(1, 3)$, the map $X \mapsto AX$ preserves $H_{R_H}^3$, d_H , geodesics, exponential and logarithm maps, parallel transport, Hausdorff distances between compact record supports, and Fréchet means whenever the mean is unique.*

Proof. The defining equation of H_{RH}^3 and the formula for d_H depend only on the Lorentz inner product and the future sheet. Elements of $\text{SO}^+(1, 3)$ preserve both. The Levi-Civita connection, geodesic exponential, logarithm, and parallel transport are functorial for isometries, so the listed constructions commute with A . Hausdorff distance and unique Fréchet means are metric constructions, hence are preserved by any isometry. $\square$

13.2 Cap-response localization balls

The compact paper supplies the conditional inverse theorem that turns calibrated modular cap responses into localized record supports. For every descended record token i and clock slice t , this particle/worldline layer consumes a certificate

$$R_i(C, t, O) = \omega_{i,O} \left(\sigma_t^{C,O}(M_{C,0,O}) \right), \quad y_{i,j} = F_j(X_i(t)) + e_{i,j},$$

on a compact source domain $\Omega \subset H_{RH}^3$, with quantitative observability

$$\alpha \bar{d}(X, Y) \leq |F(X) - F(Y)|_W \leq L \bar{d}(X, Y), \quad |e_i|_W \leq \sigma_i(t).$$

Given an ε -net and residual tolerance $\tau_i(t)$, the imported theorem emits

$$S_i(t) = B_H(\hat{X}_i(t), r_i(t)), \quad r_i(t) = R_H \left[\frac{L}{\alpha} \varepsilon + \frac{2}{\alpha} \sigma_i(t) + \frac{1}{\alpha} \tau_i(t) \right].$$

A point estimate $\hat{X}_i(t)$ is a unique finite localization output only when $\Delta_{\text{loc},i}(t) > 0$. If this gap fails, the input to the stitch theorem is an ambiguity set or support ball, not an arbitrary token-ID tie-break. Distinct record sources at a shared clock slice require the certified separation

$$d_H(\hat{X}_i, \hat{X}_k) > r_i + r_k + m_{\text{sep}}$$

for a declared positive margin m_{sep} .

13.3 Proto-record extraction and descent

A raw local proto-record is a connected component of an invariant detector scalar above a frozen threshold, together with its certified localization ball or support set, local clock interval, sector payload, and uncertainty collar. The detector scalar is built from record-algebra data, not from implementation object IDs. Extraction is chart-natural when transporting the raw field by G_{ji} and F_{ji} carries extracted components in chart i to the extracted components in chart j , up to the declared support collar.

Before temporal stitching, overlapping charts are descended. The overlap-descent relation joins two proto-records only when their transported supports lie in the same connected component on a real chart overlap and the same-component margin dominates the support collar. If distinct component separation does not dominate the collar, the descent emits an ambiguity certificate instead of choosing a label. On component-connected covers, this descent produces canonical global descended records by the usual sheaf gluing argument: cocycle-compatible local components with positive separation margins have a unique global component, and the construction is natural under atlas restriction.

13.4 Boundary-crossing germs and transported residuals

A cross-boundary candidate edge is admissible only when the two descended records meet a real patch interface in adjacent observer-clock intervals and share a common comparison chart. The interface contact must be oriented and transverse: the signed-distance interval brackets the interface, and the normal velocity has a positive lower bound. Tangencies, grazing contacts, and file or shard-name boundaries without a physical interface are rejected. If the local slab contains an interaction marker, the record is routed to the interaction solver instead of stitched by the free-propagation rule.

For an admissible pair, the certificate transports the left record into the right comparison chart and evaluates frozen residuals

$$r_t, \quad r_\partial, \quad r_x, \quad r_v, \quad r_{\text{int}},$$

for clock adjacency, boundary contact, H^3 position, transported velocity or finite difference, and interaction absence. The cost function is declared before seeing the proposed matching. Record IDs, global IDs, shard-local IDs, and stitch keys are forbidden in admissibility, costs, and tie-breaking. All H^3 position and velocity residuals are evaluated as certified interval or uncertainty-ball distances. A point-estimate residual may be printed as a diagnostic, but it cannot replace the localization radius in an admissibility or matching certificate.

Lemma 13.2 (Candidate-edge naturality). *The admissible candidate-edge set and its residual intervals are invariant under common $\text{SO}^+(1, 3)$ chart changes and under gauge-bundle trivialization changes.*

Proof. Every geometric term is computed from d_H , oriented interface data, and the common clock line, all of which are preserved by Theorem 13.1 and clock-map compatibility. Sector and gauge residuals compare transported payloads after applying the declared connector, so changing a local trivialization conjugates both sides of the comparison. The residual intervals and the resulting admissibility predicate are therefore unchanged. $\square$

13.5 Assignment gap, nonbranching paths, and ambiguity

Let L and R be the descended record sets on two adjacent time slabs. A stitch matching is a partial one-to-one assignment between L and R , augmented by declared appearance and disappearance interval costs. Its certified gap is

$$\Delta_{\text{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*),$$

where M_* is the proposed matching, $\overline{C}(M_*)$ is its upper cost bound, and $\underline{C}(M)$ is the lower cost bound for every competing matching. The localization gap Δ_{loc} and stitch-assignment gap Δ_{cert} are distinct receipts: the first certifies a unique spatial support inside one clock slice, while the second certifies a unique temporal assignment across slices.

Theorem 13.3 (Robust ID-independent stitch assignment). *If the candidate-edge set is natural, the costs are frozen and ID-independent, the proposed matching is one-to-one, and $\Delta_{\text{cert}} > 0$, then M_* is the unique certified minimum-cost stitch. The certified matching is natural under chart changes, gauge trivializations, and relabeling of implementation record IDs.*

Proof. The strict gap says every competing matching has lower cost bound above the proposed matching's upper cost bound, so no competitor can tie or beat it under any realization inside the certified intervals. Lemma 13.2 makes the candidate graph and cost intervals invariant under chart

and gauge presentation changes. Because IDs are excluded from admissibility and costs, relabeling them changes only bookkeeping fields, not the optimum. $\square$

Theorem 13.4 (Unavoidable ambiguity). *If a nontrivial permutation of the visible descended records preserves the atlas, clocks, interfaces, sector/gauge payloads, candidate graph, and cost intervals, then no natural ID-independent finite procedure can certify a unique stitch across that slab.*

Proof. A natural procedure must commute with the permutation. A unique certified output would therefore be fixed by the permutation. But the permutation sends one equally admissible stitch to a distinct equally admissible stitch. Choosing between them would introduce a label or presentation dependence. The only natural finite output is an ambiguity label. $\square$

The certified stitch graph is then the graph obtained by chaining the unique matchings across adjacent slabs. Since each matching is partial one-to-one, every connected component is a non-branching path, ray, or isolated record. These are the H^3 record-worldlines of this section.

13.6 Refinement naturality and optional Lorentzian lift

For a refinement $s \rightarrow r$, let Q_{sr} contract fine descended records to coarse records and let η_{sr} bound the cost distortion between the fine and coarse stitch problems. If the coarse certified gap obeys $\Delta_r > 2\eta_{sr}$, then the contracted fine optimum is isomorphic to the coarse optimum. This is the refinement-naturality gate: a cross-boundary stitch is not paper-grade until the coarse/fine contraction agrees within the declared margin.

The H^3 certificate is spatial and observer-record-level. A Lorentzian worldline lift is a separate optional criterion requiring lapse/shift data, a timelike speed margin, and compatibility with the emergent four-dimensional chart. Without that lift, the certificate proves record-token continuation in the H^3 observer chart, not a full spacetime geodesic theorem.

Theorem 13.5 (Certified H3 record-worldline stitching). *Given a valid observer-facing H^3 chart atlas, a common operational clock atlas, and a passing cap-response localization certificate for every descended record token and clock slice, chart-natural proto-record extraction, overlap descent with positive localization-ball separation margins, real transverse interface-crossing germs, sector and gauge transport continuity, a frozen ID-independent one-to-one assignment with $\Delta_{\text{cert}} > 0$, no interaction marker in the free-propagation slab, and a passing coarse/fine contraction gate, the emitted stitch graph is a unique natural nonbranching H^3 record-worldline graph. If any positive localization gap, separation margin, or assignment gap is absent, the natural output is AMBIGUOUS; if an interaction marker is present, the output is INTERACTION REQUIRED; if atlas, interface, localization, or transport gates fail, the stitch is rejected.*

Proof. The atlas, localization, and extraction gates make local proto-records presentation-independent. The positive Δ_{loc} gate licenses point outputs; otherwise the procedure carries support balls or ambiguity sets forward. Descent turns overlap-compatible local components into descended records before any temporal decision is made. The interface and transport gates define a natural candidate-edge graph by Lemma 13.2. The strict assignment gap gives a unique ID-independent matching by Theorem 13.3. Chaining partial one-to-one matchings gives nonbranching components, and the refinement gate makes the result stable under the declared coarse/fine contraction. The alternative statuses are exactly the negations of these gates, with Theorem 13.4 forcing ambiguity when the visible data do not separate a unique continuation. $\square$

13.7 Scope boundary of the measurement chapter

This whole chapter has to preserve one final distinction. The fixed-cutoff measurement interface is theorem-bearing on the microphysics surface: central record algebra, Born probabilities, Lüders conditioning, repeated-read stability, and the stated two-wing Bell/CHSH theorem stack are written there as fixed-cutoff statements. What is *not* proved at the same level is the stronger global observer-continuation or strange-loop proposal sometimes discussed in broader OPH notes. This paper should therefore use the fixed-cutoff measurement package directly, while leaving the larger metaphysical closure proposal out of the main technical chain.

14 Discussion: Matter, Antimatter, Supersymmetry, and Continuation Questions

This section is intentionally not written in the same voice as the structural and quantitative sections above. The OPH theorem surface distinguishes sharply between recovered-core results, quantitative-closure branches, secondary quantitative branches, and continuations. Matter versus antimatter, supersymmetry, and several adjacent questions lie on that boundary. They are too important to omit from a particle-spectrum paper, but they are not part of the closed prediction surface in the same way that the bosonic and quark rows are.

14.1 Matter and antimatter

The Standard Model structural branch fixes the chiral gauge architecture within which matter and antimatter are defined. Once a chiral fermion representation is present, its conjugate representation gives the corresponding antiparticle content. In that limited sense, OPH explains why matter and antimatter both exist: they are paired by the realized gauge and chiral structure of the low-energy branch.

What is *not* structurally closed is the cosmological asymmetry between them. The integrated continuation notes record a baryogenesis continuation tied to the realized $\mathbb{Z}_6$ quotient and the electroweak topological channel. In that continuation, the entropy deficit of the $\mathbb{Z}_6$ quotient gives a suppression factor

$$\varepsilon = e^{-\ln 6} = \frac{1}{6},$$

and the same notes compare powers of that factor with the 12-fermion electroweak baryon-violating channel. But that comparison is only a suppression-counting heuristic. It is not a dynamical derivation of the observed baryon asymmetry.

But the observer-side claim notes are explicit about the scope of that estimate: the baryon asymmetry scale is a Phase-III continuation, and suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism. The declared derivation also does not expose a baryogenesis output row. So this paper can report the available OPH continuation idea, but it should not present a sharp matter/antimatter abundance prediction as if it were a closed theorem or an emitted output.

The required ingredients are specific: a concrete out-of-equilibrium epoch, justified defect/sphaleron production dynamics, derived CP-odd source terms, transport and washout control, and a freeze-out computation of the final asymmetry. The available corpus does not have a baryogenesis theorem package. The same continuation also notes that CP-odd overlap data are generically available because central overlap phases are sent to their inverses under CP, but that observation is not a baryogenesis theorem.

14.2 Why the declared branch does not require supersymmetry

The D10 appendix below contains the dedicated gauge-coupling quantitative-closure discussion unification. Its main point is not that supersymmetry has been mathematically ruled out in every possible extension of the program. Its point is narrower and more relevant to this paper: the declared branch does not *need* a supersymmetric partner spectrum in order to explain why unification-like running might appear.

At one loop, the integrated D10 appendix records the familiar fact that Standard Model beta-function coefficients do not produce successful naive unification, whereas MSSM-like coefficients do. The OPH continuation idea is then that edge-sector heat-kernel weights can shift the effective beta-function coefficients in an MSSM-like direction through geometric or entropic sector multiplicity instead of through an actual low-energy superpartner spectrum. In that reading, “unification-like behavior” and “a supersymmetric particle zoo” come apart.

That continuation is narrower than a derived unification theorem. The appendix records one calibration branch in which Peter–Weyl multiplicities, the printed one-loop running frame, the threshold conventions, and an additional fermionic-grading restriction together produce an MSSM-like benchmark shift. It does not prove that those effective loop multiplicities, statistics restrictions, and decoupling conventions are uniquely forced by OPH edge sectors alone.

For this paper, the correct conclusion is therefore scope-limited and modest:

1. the derivation reconstructs the realized Standard Model branch without having to introduce a supersymmetric partner for every known field;
2. the integrated D10 appendix contains a continuation-level argument that some unification-style running features could arise from edge-sector structure instead of MSSM particle content;
3. the beta-shift package itself sits on explicit D10 calibration assumptions instead of a closed edge-sector theorem;
4. none of this is a universal theorem that supersymmetry is impossible.

So if the paper asks “why is there no supersymmetry in the derived spectrum?”, the best technical answer is: because the realized branch stops at the Standard Model content, and the existing unification discussion is trying to reproduce the relevant running behavior through a separate D10 calibration package, without extending the particle content to a full supersymmetric multiplet structure.

14.3 Three Generations And Flavor Closure

The structural Standard Model branch fixes $N_g = 3$ on the realized MAR-admissible branch. This result is separate from the harder flavor closure problem. Three generations are structurally recovered; the full flavor dictionary, excitation map, and CKM closure are not emitted on this theorem surface, while the charged source landing from P to physical charged data is a corpus-limited no-go and the neutrino lane has no source-closed flavor prediction. Its weighted-cycle candidate fails the NuFIT 6.1 correlated profile [9]. The flavor chapter and the later family chapters therefore spend substantial space on constructive object boundaries as well as final numbers.

14.4 Ordinary matter and the hadron backend boundary

A second distinction belongs in discussion form instead of in the results sections: the difference between “elementary rows exist” and “ordinary matter is fully derived.” Ordinary visible matter is

dominated by hadrons, especially protons and neutrons. The OPH derivation has a hadron pipeline, with source-only hadron masses gated by a production backend export. The paper emits the eligible elementary rows and retains the weighted-cycle coordinates only as a rejected comparison record, while the full observed matter spectrum is not emitted from a source-only OPH backend.

14.5 How to read these continuation questions

The reader should therefore interpret this whole discussion section with the same rule used by *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5]. Charged-lepton source landing, fuller flavor-labeled neutrino, and source-only hadron mass theorems sit beyond the closed structural core; failure in those continuation branches would not undo the recovered-core gauge and gravity chain or erase the reported bosonic and quark rows.

15 Conclusion

The particle-spectrum derivation is nontrivial and much more concrete than a qualitative aspiration. The structural theorem surface fixes the realized Standard Model branch and the exact massless carrier sectors. The completion pipeline closes or scopes the structural massless carriers, the hierarchy/naturality bridge on the selected P/N branch, the Higgs/top row on the declared D10/D11 surface, and selected-class quarks on f_P . The neutrino lane records an isotropic no-go, a rejected target-informed weighted-cycle candidate, and open source-basis, mass-label, and operator gates. The ledger also marks four separate row classes: the candidate P -closure source audit, the electroweak W/Z validation row, the charged-lepton absolute-mass landing, and the hadron backend plus empirical closure split. The W/Z row is a frozen compare-only adapter with its own public validation records. The Ward-projected Thomson lane records the source/root audit, the mixed-provenance root-plus-comparison-pixel diagnostic, and the measured endpoint as separate rows on the declared P -closure surface. In the charged lane the theorem surface carries an exact same-family witness, a determinant-line lift, and an algebraic mass readout from theorem-grade $A_{\text{ch}}(P)$, while the source landing from the D10 quantitative descendants of P to physical charged data is closed as a corpus-limited no-go. Source-only hadron masses require a working OPH hadron backend. The declared empirical route would use a separate $e^+e^- \rightarrow$ hadrons payload class; no integrated endpoint payload is emitted here.

That accounting belongs on the page because it is part of the derivation. The paper puts the structural results, the fixed-point fine-structure value, the emitted quantitative rows, and the claim-boundary ledger onto one common surface. The bosonic P -driven trunk is explicit. The local extension surface identifies the shared edge entropy with the D10 nonabelian sum on the lifted product presentation of the realized quotient branch, and the same surface fixes that scalar to $P/4$. The local/global hierarchy bridge adds the exact capacity value $N_{\text{CRC}}^{\text{EW}}$, the zero bridge residual, the 12-port screen sieve, the 24-slot oriented repair register, and $\epsilon_H = 0$. The particle spectrum is therefore organized by one local screen scale instead of by a list of unrelated particle inputs. The claim ledger records scope boundaries for the weak validation pair, charged-lepton witnesses, direct-top auxiliary codomain, strong-CP phase boundary, neutrino comparison tension, source-only hadron backend rows, and empirical hadron closure rows.

A D10 Heat-Kernel Calibration Supplement

This appendix carries the D10 compact-group heat-kernel items needed by the particle surface. The theorem-bearing leaf for the same-overlap thermal/Casimir law sits on the screen-microphysics paper surface at fixed cutoff. What is recorded here is the compact-group / Peter–Weyl lift and beta-shift bookkeeping used by the particle-side calibration branch. The public D10 claim tier is explicitly Phase II rather than recovered-core Phase I. In particular, this appendix does not prove that OPH derives the physical one-loop beta coefficients from edge sectors alone.

A.1 $\mathcal{R}_U$ Unified-Coupling Certificate v0.2

The $\mathcal{R}_U$ certificate is the D10 gate needed by the hierarchy discussion in the synthesis paper. It certifies the unified diffusion coupling used in the electroweak transmutation factor

$$\frac{v}{E_\star} = P^{-1/2} \exp \left[-\frac{2\pi}{4\alpha_U(P)} \right].$$

It does not certify the full clock branch

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

The full $\mathcal{R}_\gamma$ stack requires source-only electromagnetic endpoint, charged-lepton absolute-scale, cesium nuclear, and atomic spectral enclosures. The v0.2 package is a numerical witness for $\mathcal{R}_U$, with formal promotion gated by outward-rounded interval arithmetic.

Frozen source packet. The public-pixel branch uses

$$P_C = \frac{1.6309682094039593248792798477826489413359828516279250606661}{507533907793398933432}.$$

The source-audit branch uses

$$P_{\text{cand}} = 1.63097209569432901817967892561191884270169.$$

The finite heat-kernel cutoffs are

$$N_2 = 128, \quad N_3 = 64.$$

The one-loop running packet is

$$(b_1, b_2, b_3) = \left(\frac{33}{5}, 1, -3 \right), \quad N_c = 3, \quad \beta_{\text{EW}} = N_c + 1 = 4.$$

This packet is frozen in the certificate. It is not silently derived in this appendix from the recovered OPH core.

Definition A.1 (No-hidden-free-variable rule for $\mathcal{R}_U$). *The following objects must be fixed before comparison with electroweak or gravity data:*

$$(b_1, b_2, b_3), \quad N_2, \quad N_3, \quad M_U(P), \quad E_{\text{cell}}(P), \quad \beta_{\text{EW}}, \quad \text{Casimir normalization}, \quad \text{matching scheme}.$$

Changing any of them after comparison with M_W , M_Z , $\alpha_s(M_Z)$, low-energy electroweak couplings, or G_{exp} demotes the row to calibration.

Dimensionless source map. Set $E_\star = 1$ for the dimensionless calculation. For a trial coupling a , define

$$M_U(P) = e^{-2\pi} P^{1/6}, \quad E_{\text{cell}}(P) = P^{-1/2},$$

$$v(P, a) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{4a}\right].$$

For $i = 1, 2, 3$,

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \quad \alpha_Y(\mu; P, a) = \frac{3}{5} \alpha_1(\mu; P, a).$$

The source Z -scale is the positive solution of

$$\mu = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu; P, a) + 4\pi\alpha_Y(\mu; P, a)}.$$

For $\text{SU}(2)$, with $j = n/2$ and $0 \leq n \leq N_2$,

$$d_j = 2j + 1, \quad C_2(j) = j(j + 1).$$

For $\text{SU}(3)$, with $0 \leq p, q \leq N_3$,

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}, \quad C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

The group sums are

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \quad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

Define

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \quad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a),$$

and

$$\Phi_U(P, a) = \bar{\ell}_{\text{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\text{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The certificate value is the zero $\Phi_U(P, \alpha_U(P)) = 0$.

Numerical witness. For $P = P_C$, the v0.2 checker emits

$$\alpha_U(P_C) = 0.041124336195630495,$$

$$\alpha_U(P_C)^{-1} = 24.316501918546496.$$

The corresponding dimensionless hierarchy data are

$$\frac{M_U(P_C)}{E_\star} = 0.0020260720012100037,$$

$$\frac{v(P_C)}{E_\star} = 2.0199803239725553 \times 10^{-17},$$

$$\frac{\mu_Z(P_C)}{E_\star} = 7.502403238986022 \times 10^{-18}.$$

The emitted couplings at μ_Z are

$$\begin{aligned}\alpha_1 &= 0.016885708038988166, \\ \alpha_Y &= 0.010131424823392899, \\ \alpha_2 &= 0.033777889448908055, \\ \alpha_3 &= 0.11833602747445048.\end{aligned}$$

The heat-kernel parameters and entropy readouts are

$$\begin{aligned}t_2 &= 1.3334976254578108, \\ t_3 &= 4.6717191102770705, \\ \bar{\ell}_{\text{SU}(2)} &= 0.39488144681077636, \\ \bar{\ell}_{\text{SU}(3)} &= 0.012860605540213493.\end{aligned}$$

Thus

$$\begin{aligned}\bar{\ell}_{\text{SU}(2)} + \bar{\ell}_{\text{SU}(3)} &= 0.40774205235098987, \\ P_C/4 &= 0.4077420523509898,\end{aligned}$$

and

$$\Phi_U(P_C, 0.041124336195630495) = 5.55 \times 10^{-17}.$$

A centered derivative check gives

$$\partial_a \Phi_U(P_C, a) \simeq -10.990524683118785.$$

With $\delta = 10^{-6}$, the bracket signs are

$$\begin{aligned}\Phi_U(P_C, \alpha_U - \delta) &= 1.0990748697592423 \times 10^{-5}, \\ \Phi_U(P_C, \alpha_U + \delta) &= -1.0990300680135956 \times 10^{-5}.\end{aligned}$$

For the source-audit pixel,

$$\begin{aligned}\alpha_U(P_{\text{cand}}) &= 0.04112424744557487, \\ \alpha_U(P_{\text{cand}})^{-1} &= 24.316554395881205, \\ \frac{v(P_{\text{cand}})}{E_\star} &= 2.0198114150099223 \times 10^{-17}, \\ \frac{\mu_Z(P_{\text{cand}})}{E_\star} &= 7.501767385088398 \times 10^{-18}.\end{aligned}$$

This branch is the one to cite when avoiding the measured Thomson endpoint in the upstream pixel solve.

Theorem A.2 (Electroweak unified-coupling numerical witness). *Assume the frozen source map, running packet, heat-kernel cutoff policy, and no-hidden-free-variable rule above. Let*

$$I_U = [0.041123336195630494, 0.041125336195630496].$$

If outward-rounded interval arithmetic verifies

$$0 \in \Phi_U(P_C, I_U), \quad 0 \notin \partial_a \Phi_U(P_C, I_U),$$

then there is a unique $\alpha_U(P_C) \in I_U$ satisfying

$$\Phi_U(P_C, \alpha_U(P_C)) = 0.$$

Moreover, the dependency graph of this zero contains no measured G , no Planck unit formed using measured G , no measured Λ , no measured M_Z , no measured M_W , and no measured low-energy gauge coupling.

Proof. Continuity follows because the running couplings, self-consistent Z -scale solution, and finite heat-kernel sums are continuous on the declared positive interval. The endpoint sign change gives existence by the intermediate value theorem. The derivative exclusion gives uniqueness. The dependency statement follows by inspection of the frozen source map: the residual is built only from P_C , the frozen running packet, the self-consistent Z -scale equation, and finite compact-group representation sums. $\square$

Precision policy. The exponential map makes downstream gravity displays sensitive to α_U :

$$\frac{\partial \log G}{\partial \alpha_U} \simeq \frac{\pi}{\alpha_U^2} \approx 1.86 \times 10^3.$$

Therefore the number of digits printed for any downstream G or ε_{Cs} row may not exceed the interval precision certified by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

The v0.2 witness supports an electroweak hierarchy certificate. It does not support a 52-digit source-only gravity prediction.

A.2 One-Loop Calibration Frame

At one loop, if couplings unify at (M_U, α_U) ,

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}.$$

On the D10 branch these are the branch values $(M_U(P), \alpha_U(P))$ emitted by the forward D10 solve; they are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2} (A_1 - A_2),$$

and the corresponding consistency relation for the D10 lane is

$$A_3^{\text{pred}} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH derives a supersymmetric UV spectrum.

A.3 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
SM	$(41/10, -19/6, -7)$	≈ 0.071
MSSM	$(33/5, 1, -3)$	≈ 0.116
Observed		0.1179 ± 0.0010

At one loop, the printed MSSM-style coefficients reproduce the observed closure far better than the plain SM coefficients. On this D10 branch this is a benchmark matched by the D10 calibration branch, not a theorem that OPH derives a supersymmetric UV spectrum.

A.4 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt plus the bi-invariant compact-group package yield the familiar $d_R e^{-tC_2(R)}$ law once the Peter–Weyl lift is supplied; they are not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

Theorem A.3 (Heat-kernel edge-sector weights). *Under MaxEnt with bi-invariant constraints on a compact Lie group G , the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R e^{-tC_2(R)},$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Lemma A.4 (Bi-invariant operators). *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on G has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

where Δ_{G_i} is the Laplace–Beltrami operator on the factor G_i .

Remark A.5. *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

A.5 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity d_R in p_R , but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R d_R T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor a . This is the exact mathematical point on the D10 lane where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running is a declared Phase II benchmark, not a promoted theorem-level MSSM spectrum claim.

The branch boundary for theorem-level unification closure is explicit. One requires a derived refinement-limit identification of which transportable sectors actually contribute to the running carrier, a derived statistics/grading rule rather than an imposed fermionic-grading restriction, controlled threshold and decoupling conventions on that same carrier, a derived representation content/truncation rule for the sectors retained in the comparison, and a proof that the effective loop multiplicities entering Δb_a are exactly the ones used in this calibration package rather than nearby alternatives. The declared runtime surface does not emit the full RG/matching/threshold/scheme packet: scheme lock, threshold map, beta provenance, and interval composition are separate certificate requirements.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward D10 solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \quad \text{vs MSSM} \quad (2.50, 4.17, 4.00).$$

With the additional D10 calibration assumption of a fermionic-grading restriction to half-integer $SU(2)$ sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

B H3 Worldline-Stitch Certificate

This appendix records the finite evidence-facing certificate for Theorem 13.5. It is separate from the D10 heat-kernel calibration package above. A passing certificate is a record-continuation statement in a declared H^3 observer chart, not a particle-species, mass, charge, or scattering-amplitude theorem.

The certificate payload has the following required blocks.

1. **Atlas.** It declares the hyperboloid model $H_{RH}^3 \subset \mathbb{R}^{1,3}$, the Lorentz signature $(-+++)$, the curvature radius R_H , chart domains, and chart transitions $G_{ji} \in SO^+(1,3)$. Transition residuals must certify Lorentz inner-product preservation, orientation, and future-sheet preservation.
2. **Clock map.** Each local record interval is mapped to a common comparison-time line. The observer-time adjacency margin must dominate the declared clock uncertainty.

3. **Chart-natural extraction.** Raw local records are connected components of a frozen detector scalar on the H^3 chart. The detector scalar, thresholds, support collars, and chart-naturality residuals are included. Implementation record IDs are not inputs.
4. **Overlap descent.** Records are first descended across genuine chart overlaps. The payload records same-component join margins, distinct-component separation margins, support errors, and triple-overlap cocycle residuals.
5. **Interface crossing.** A candidate stitch must cross a real interface in adjacent clock intervals. It includes oriented signed-distance margins and a positive lower bound on normal velocity. Tangential, grazing, and file-boundary-only contacts are rejected.
6. **Transport.** The candidate edge is evaluated in a common chart after sector and gauge transport. The atlas holonomy comparison is separate from the gauge-holonomy comparison.
7. **Assignment.** The matching is one-to-one with appearance/disappearance interval costs. The proposed winner has an upper cost bound, every competitor has a lower cost bound, and

$$\Delta_{\text{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*) > 0.$$

Record IDs, global IDs, stitch keys, and shard-local IDs are forbidden in admissibility, costs, and tie-breaking.

8. **Refinement.** A coarse/fine pair supplies Q_{sr} , a distortion bound η_{sr} , and a contracted-graph isomorphism check. The coarse gap must satisfy $\Delta_r > 2\eta_{sr}$.
9. **Interaction firewall.** A free-propagation slab is required. If an interaction marker is present, the terminal label is `H3_INTERACTION_REQUIRED` and the event is handed to the interaction solver.

The allowed terminal statuses are

`H3_STITCH_CERTIFIED`, `H3_STITCH_AMBIGUOUS`, `H3_STITCH_REJECTED`,
`H3_INTERACTION_REQUIRED`, `H3_ATLAS_INVALID`, `H3_CERTIFICATE_INCOMPLETE`.

`H3_STITCH_CERTIFIED` is emitted only when every block above passes and the robust assignment gap is positive. Equal-cost matchings, preserved nontrivial permutations of the visible data, or coarse/fine gaps below margin force `H3_STITCH_AMBIGUOUS`. Missing fields force `H3_CERTIFICATE_INCOMPLETE`. Failed atlas, interface, metric, or transport gates force rejection or atlas invalidity. In particular, Euclidean distances on `h3SpatialPoint` coordinates are not certificate distances; the certificate distance is the hyperboloid geodesic d_H from Section 13.

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Sector	Support status	Derivation stage	Public output(s) / exact sidecar(s)	Caveat / theorem boundary
Structural carriers	structural	gauge/gravity structural theorems	exact $m_\gamma = m_g = m_{\text{grav}} = 0$	theorem-level structural exactness; no particle-side caveat
Electroweak bosons	benchmark adapter	electroweak branch plus frozen benchmark adapter	$M_W = 80.377 \text{ GeV}$, $M_Z = 91.18797809193725 \text{ GeV}$	benchmark check beneath the declared P -to-fine-structure arithmetic bridge; public records are source spectral measure payload, same-scheme remainder, and interval certificate
Electroweak hierarchy/naturality	selected-branch theorem	local $P \rightarrow \alpha_U \rightarrow v/E_*$ hierarchy lane → 12-port screen sieve → 24-slot oriented repair register → exact capacity bridge and RG/Higgs square	$N_{\text{CRC}}^{\text{EW}} = 3.5323546226929906511 \dots \times 10^{122}$, $\mathcal{B}_{\text{EW}} = 0$, $\epsilon_H = 0$	Supports the hierarchy and Higgs naturality rows on the selected branch. The public Thomson endpoint, theorem-level W/Z status, charged-lepton absolute masses, first-principles hadrons, Strong CP, and full SI G remain separate surfaces. Higgs row is conditional on the declared surface.
Higgs/top stage	exact first-principles split theorem on the declared running, matching, and threshold surface + benchmark exact sidecar	gauge core followed by the target-free repair chart, the shared scalar $\rho_{HT} = \log(1 + \tau_{2,\text{tree}}^{\text{exact}})$, the first-principles split selectors, and the Jacobian readout	public numeric output $m_H = 125.1995304097179 \text{ GeV}$; companion top coordinate on the same surface; exact inverse-sidecar pair on the Jacobian	The companion top coordinate is not a separate public top-mass prediction row. The inverse pair is a benchmark check. The auxiliary direct-top PDG row is a distinct benchmark codomain with a corpus-limited no-go boundary.
Quark family	selected-frame conditional support + exact supporting surfaces	shared excitation dictionary → selected public quark frame f_P → representative-independent sigma descent → missing source sigma selector → affine mean law → ordered three-point readout → exact forward construction	selected-class quark/Yukawa support surface; target-anchored running sextet withheld from public prediction tables	selected-frame conditional support only. The exact sextet matches the official PDG 2025 API running-quark surface because the physical sigma datum is target-derived. It is therefore an audit witness, not a public source-only prediction. The auxiliary direct-top PDG row is a benchmark check with a corpus-limited no-go boundary. The stronger global classification of public quark frames is not emitted. This paper does not derive θ_{QCD} , does not emit physical $\hat{\theta}$, and does not prove that the physical strong-CP phase vanishes. Supporting exact surfaces use same-family and restricted common-refinement transport-frame checks. The separate target-free mass bridge closes only on the emitted mass ray. No global classification of all quark frame classes is claimed
Charged leptons	continuation + exact sidecar witness	shared excitation dictionary → ordered charged carrier → exact centered readback → determinant-line lift on theorem-level physical charged data	exact centered readback; same-family target-anchored charged triple withheld from public prediction tables	public charged masses are not emitted from P . The theorem lane does not emit a theorem-level sector-isolated charged determinant exponent vector. It does not attach a source-side determinant character to the physical charged determinant line. The determinant-line lift and algebraic mass readout apply only on theorem-level physical charged data
Neutrinos	target-informed continuation candidate + auxiliary checks	template family transport → same-label scalar certificate → weighted-cycle candidate → compare-only bridge and absolute attachment → shared-basis identity	frozen scale-free candidate and PMNS/Majorana comparison rows; absolute mass attachments withheld from public prediction tables	exact linear algebra conditional on the declared template and selectors. The candidate fails the NuFIT 6.1 correlated $(\sin^2 \theta_{23}, \delta_{\text{CP}})$ profile, and neither the scale-free point nor the bridge invariant has prediction or theorem status [9]
Hadrons	strong-binding construction absent; empirical closure policy emitted	OPH-QCD backend contract, including quotient ensemble, source QCD parameter map, Ward current ledger, spectral exports, and empirical $e^+e^- \rightarrow$ hadrons payload schema	no first-principles hadron masses; empirical hadron closure rows are separate from first-principles OPH rows	First-principles hadron prediction requires a working OPH hadron construction, such as GLOBE/Echosahedron, with the Ward-projected two-current spectral measure, higher-point/transition spectral sectors, same-scheme remainder, and systematics. The empirical closure surface uses separate $e^+e^- \rightarrow$ hadrons input and cannot upgrade the first-principles theorem
H3 record-worldline stitching	conditional certificate theorem	declared H_R^3 chart atlas and observer clock atlas → cap-response localization certificate for each descended token and clock slice → localization balls with positive Δ_{loc} for point outputs → overlap descent → real transverse interface-crossing germs → common-chart sector/gauge transport → ID-independent one-to-one assignment gap → coarse/fine contraction check	nonbranching localized record-token paths, rays, or isolated records when both localization and stitch certificates pass; otherwise AMBIGUOUS, REJECTED, or INTERACTION REQUIRED	This is a continuation theorem for observer-visible records in H^3 , not a derivation of particle species, masses, gauge charges, or scattering amplitudes. Repeated implementation IDs, nearest-neighbor fits, pre-existing coordinates without response inversion, and file-boundary coincidences are not admissible evidence.

Lane	Exact output(s)	Exact chain on the paper surface	Caveat
Structural carriers	$m_\gamma = m_g = m_{\text{grav}} = 0$	axioms $\rightarrow$ realized electromagnetic/color/dynamical-metric carrier skeleton $\rightarrow$ symmetry-protected zeros	theorem-level structural exactness
Electroweak exact sidecar	exact frozen W/Z pair	axioms $\rightarrow$ quantitative-closure chain $\rightarrow$ exact frozen repair pair	exact only on the frozen authoritative repair surface; benchmark check beneath the target-free theorem
Hierarchy/naturality bridge	exact bridge residual and zero Higgs naturality defect	axioms $\rightarrow P$ -fixed source branch $\rightarrow N_{\text{CRC}}^{\text{EW}}$ bridge fixed point $\rightarrow$ screen sieve and 24-slot repair register $\rightarrow$ source-to-Higgs settled-form square	selected-branch theorem for the hierarchy and naturality rows; it does not support the W/Z , Thomson endpoint, charged-lepton, hadron, Strong CP, or SI G rows
Higgs/top exact sidecar	exact Higgs/top inverse pair	axioms $\rightarrow$ gauge core $\rightarrow$ Jacobian $\rightarrow$ exact inverse slice	benchmark inverse slice; the public numeric row is the conditional Higgs candidate from the first-principles split theorem on the declared surface. The top coordinate on that surface is not promoted as a separate public top-mass prediction row
Charged exact witness	exact same-family charged triple withheld	axioms $\rightarrow$ shared excitation dictionary $\rightarrow$ ordered charged carrier $\rightarrow$ exact centered readback $\rightarrow$ closed quadratic readout theorem $\rightarrow$ same-family exact witness	same-family target-anchored audit witness only; theorem lane carries exact centered readback plus the closed common-shift no-go. The available derivation has a no-go boundary: a theorem-level $\widehat{C}_e$ lift emitting the physical scalar $\mu_{\text{phys}}(Y_e)$ is not part of the emitted theorem surface
Quark selected-frame conditional support	target-anchored running quark sextet withheld; explicit forward Yukawa support surface retained on f_P	axioms $\rightarrow$ shared excitation dictionary $\rightarrow$ selected public quark frame $f_P \rightarrow$ representative-independent sigma descent $\rightarrow$ missing source sigma selector $\rightarrow$ affine mean law $\rightarrow$ ordered three-point readout $\rightarrow$ exact forward construction	selected-frame audit/support closure on the public physical quark frame chosen by P , conditional on source sigma; the sextet matches the official PDG 2025 API running-quark surface because the physical sigma datum is target-derived; the auxiliary direct-top PDG row is a benchmark check with a corpus-limited no-go boundary; the stronger global classification of public quark frames is not emitted; no global classification of all quark frame classes is claimed
Neutrino candidate	frozen scale-free comparison point; absolute mass attachments withheld	template family transport $\rightarrow$ same-label scalar certificate $\rightarrow$ target-informed weighted-cycle law $\rightarrow$ compare-only bridge and absolute attachment $\rightarrow$ shared-basis identity	rejected by the NuFIT 6.1 correlated profile; source closure, historical blindness, basis orientation, and absolute normalization all fail promotion gates [9]

Carrier	Claim tier	OPH output	Physical role	Why the mass is zero
Photon	structural	0 GeV	electromagnetic force carrier	gauge redundancy forbids a hard mass
Gluons	structural	0 GeV	color-force carriers	color-gauge redundancy forbids a hard mass
Graviton	structural	0 GeV	spin-2 metric carrier	diffeomorphism redundancy forbids a hard mass

Particle	Stage	Claim tier	Reported value
W boson	D10 electroweak branch	compare-only frozen adapter	80.377 GeV
Z boson	D10 electroweak branch	compare-only frozen adapter	91.18797809193725 GeV
Higgs boson	Higgs/top critical stage	quantitative theorem	125.1995304097179 GeV
Top quark	Higgs/top critical stage	companion coordinate on declared surface; direct-top auxiliary codomain no-go	not a separate public top-mass prediction row

Quark	Claim tier	Public value	Derivation stage
Up quark	conditional on source sigma	withheld	selected public quark frame class f_P
Down quark	conditional on source sigma	withheld	selected public quark frame class f_P
Strange quark	conditional on source sigma	withheld	selected public quark frame class f_P
Charm quark	conditional on source sigma	withheld	selected public quark frame class f_P
Bottom quark	conditional on source sigma	withheld	selected public quark frame class f_P
Top quark	conditional on source sigma	withheld	selected public quark frame class f_P